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PREDICTING AND EXPLAINING TRAFFIC COLLISION SEVERITY IN TORONTO USING EXPLAINABLE MACHINE LEARNING

by

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ABSTRACT

Traffic collisions remain a critical public safety concern in Toronto, and while the City publishes detailed historical collision records, most existing analyses remain descriptive rather than predictive or explainable. This research develops and evaluates a machine learning framework to predict collision severity Minor, Major, or Fatal from time, location, weather, and roadway-context features, using 677,056 cleaned collision records (2014–2026) merged with hourly weather data from the Government of Canada Historical Climate Database. Five classifiers (Logistic Regression, Random Forest, XGBoost, Gradient Boosting Machines, and a Multi-Layer Perceptron) were compared via 5-fold cross-validation, after correcting two methodological issues identified during development: outcome-correlated involvement-flag features that constituted data leakage, and a categorical-encoding error in which SMOTE-NC resampled one-hot encoded temporal features as independent binary columns rather than as single categorical variables. After both corrections, Fatal-class Precision-Recall AUC remained low across all five models (0.0010–0.0017), values that are close in absolute terms to the approximate Fatal-prevalence reference level of 0.0009. XGBoost achieved the highest F1 Macro (0.338) and Accuracy, making it the strongest overall three-class model under the aggregate ranking criterion, while Random Forest achieved the highest Fatal-class PR-AUC (0.0017), making it the strongest model on the Fatal-specific discrimination metric. The broadly weak Fatal-specific results across several model families support the interpretation that the current predictors provide limited Fatal-class separability under the modelling setup used here, while not ruling out improvements from alternative predictors, model classes, or evaluation designs. SHAP analysis of the corrected model highlighted `is_weekend`, `OCC_YEAR`, and location features as important model dependencies, while DBSCAN spatial clustering identified 36 Fatal-collision clusters across Toronto. Because most Fatal observations were spatially isolated under the chosen DBSCAN parameters, these clusters are interpreted descriptively as a supplementary spatial prioritization layer for further investigation rather than as a comprehensive risk map.

Key words: Traffic collision severity, machine learning, class imbalance, SHAP explainability, data leakage, geospatial analysis

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Contents

1 Introduction

Traffic collisions are a persistent public safety and public health problem in large urban centres. In Toronto alone, hundreds of thousands of collisions are recorded each year, ranging from minor property-damage incidents to collisions resulting in serious injury or death. The City of Toronto has committed to Vision Zero, a road safety strategy aimed at eliminating traffic-related fatalities and serious injuries. Achieving this goal requires more than descriptive statistics on where and when collisions occur it requires the ability to anticipate which collisions are likely to be severe, and to explain why, so that limited safety resources can be directed where they will have the greatest impact.

Most existing analyses of Toronto's collision data are descriptive: they summarize historical patterns by location, time, or road user type, but do not attempt to predict severity for new or hypothetical collisions, nor do they explain which factors drive a model's predictions in a way that is trustworthy for policy use. Machine learning offers a path toward both predictive and explainable analysis, but applying it responsibly to a safety-critical, severely imbalanced outcome where the Fatal class represents a small fraction of one percent of all records requires careful attention to class imbalance, model evaluation, and the risk of data leakage from features that are, in effect, recorded as a consequence of the outcome rather than as a genuine predictor of it.

This research asks two questions. First, can a machine learning model predict the severity of a Toronto traffic collision Minor, Major, or Fatal using only information that would plausibly be available independent of the collision's outcome (time, location, weather, and roadway context)? Second, which available features does the fitted model rely on when producing its predictions, and can spatial analysis of confirmed Fatal collisions identify geographic concentrations that warrant further road-safety investigation?

This project makes the following contributions:

- A complete, reproducible machine learning pipeline for Toronto collision severity prediction, integrating the City's full Motor Vehicle Collisions dataset (not the smaller Killed-or-Seriously-Injured subset used in several prior studies) with hourly weather observations from Environment and Climate Change Canada.
- A systematic data-leakage audit of the involvement-flag features in this dataset, showing that several commonly-used features are populated in a manner correlated with the severity label, and a corrected feature set with these features removed.
- A methodologically corrected treatment of class imbalance using SMOTE-NC, which respects the categorical structure of one-hot encoded temporal features, together with a systematic comparison against four alternative balancing strategies.
- A cross-validated, full-dataset model comparison across five classifier families, reported as mean $\pm$ standard deviation rather than a single train/test split, together with PR-AUC as a metric appropriate to the severity of the class imbalance.

- An integrated explainability and geospatial framework, combining SHAP-based interpretation of the corrected model with DBSCAN-based spatial clustering of fatal collisions, intended to support Vision Zero prioritization.

The remainder of this report is organized as follows. Section 2 reviews the relevant literature and identifies the research gaps addressed by this study. Section 3 describes the datasets, preprocessing, feature engineering, and exploratory analysis. Section 4 presents the modelling methodology, including class-imbalance handling, model selection, evaluation, explainability, and geospatial analysis. Section 5 reports the experimental results. Section 6 discusses and interprets these results, including their practical implications and their relationship to the literature reviewed in Section 2. Section 7 concludes, and Section 8 states the limitations and threats to validity of this study.

2 Literature Review

Traffic collision severity prediction has attracted growing attention in transportation research, particularly as large-scale government datasets and advanced machine learning techniques have become widely accessible. This section synthesizes eight studies that inform the modelling, explainability, and geospatial components of this project, organized thematically rather than study-by-study.

2.1 Ensemble Methods for Severity Classification

A recurring finding across the literature is that tree-based ensemble methods outperform linear baselines on structured collision data. Chen and Wang [2] compared Logistic Regression, Decision Trees, Random Forest, and Support Vector Machines on a large U.S. national crash dataset and found that Random Forest outperformed all alternatives, with time of day, road surface condition, and vehicle count emerging as the strongest predictors; all tree-based ensembles substantially outperformed the linear baseline. Sheykhfard et al. [9] extended this comparison to Gradient Boosting Machines on Iranian rural road data, again finding that boosted trees outperformed both Logistic Regression and Neural Networks, with seatbelt usage, speed, and road curvature as the leading risk factors. Parsa et al. [6] applied XGBoost together with SHAP explainability to highway crash data for real-time crash detection and feature analysis, using SMOTE to address class imbalance in the detection task directly motivating this project's choice of XGBoost as its primary model, its use of SMOTE-family oversampling, and its pairing of a boosted-tree model with SHAP for explainability. Sameen and Pradhan [8] offer a partial counterpoint: a Recurrent Neural Network was found to be competitive with, and in some configurations to modestly outperform, a Random Forest baseline on Malaysian highway data, at substantially higher computational cost and reduced interpretability, suggesting that for moderate-sized tabular datasets, ensemble methods offer the better performance–interpretability trade-off. Taken together, this literature supports the five-model comparison Logistic Regression, Random Forest, XGBoost, Gradient Boosting, and a Multi-Layer Perceptron adopted in this study, spanning a linear baseline, two boosted-tree variants, a bagged-tree variant, and a deep learning baseline.

2.2 Environmental and Contextual Features

Beyond model choice, several studies show that enriching collision data with contextual and environmental features materially improves prediction quality. Rezapour et al. [7] analyzed injury severity in motorcycle at-fault crashes using decision tree and logistic regression models, contributing evidence that tree-based methods can outperform a linear baseline on crash-severity classification and that crash-context features beyond basic demographics carry real predictive value supporting this project's inclusion of a broad contextual feature set rather than a minimal one. Al-Ghamdi [1] took an earlier, geospatially-oriented approach, applying multinomial Logistic Regression with road type, intersection type, lighting condition, and GPS-derived spatial variables to Saudi Arabian urban collision data, finding that highway collisions and poor lighting were both associated with higher severity, and establishing Logistic Regression as a useful, interpretable if limited baseline. These studies jointly motivate this project's decision to merge Toronto collision records with hourly Environment and Climate Change Canada weather data, and to retain GPS-derived spatial and neighbourhood features in the model.

2.3 Explainability

A model's predictive accuracy is of limited practical use in a policy context if its reasoning cannot be audited. Lundberg et al. [5] introduced SHAP (SHapley Additive Explanations), grounded in cooperative game theory, and showed that TreeSHAP enables exact, consistent global and local explanations for tree ensembles in linear time, outperforming earlier interpretability methods such as LIME and permutation importance in both consistency and computational efficiency. This is the foundational justification for this project's use of SHAP, rather than a simpler feature-importance heuristic, to interpret the final XGBoost model.

2.4 Spatial Analysis

Collisions are not spatially uniform, and identifying where severe collisions cluster can provide a descriptive spatial layer for further transportation-safety investigation. Xie and Yan [10] applied Kernel Density Estimation and network-based spatial analysis to pedestrian accident data in a U.S. urban area, showing that density-based methods can identify high-risk corridors and intersection clusters without requiring the number of clusters to be specified in advance a property shared by DBSCAN, the density-based clustering algorithm adopted for the geospatial component of this study.

2.5 Research Gaps

This literature is overwhelmingly drawn from highway or rural crash data in the United States, Malaysia, Iran, and Saudi Arabia; dense urban Canadian collision data and Toronto in particular is essentially absent from the international collision-severity literature. Few of the reviewed studies combine predictive modelling, explainability, and geospatial clustering within a single pipeline, instead treating these as separate lines of inquiry. None conduct a systematic comparison of multiple class-balancing strategies (SMOTE-family variants, undersampling, and class-weighting) on the same dataset, and none integrate Canadian climate variables including snow, ice, and freezing rain, which are largely absent from the U.S.- and Asia-centric literature reviewed above as environmental predictors. Perhaps most importantly, none of the reviewed studies examine whether their engineered involvement-type features (e.g., whether a pedestrian or cyclist was involved) might be partially determined by the collision outcome itself, rather than genuinely independent of it a data-leakage risk that this project identifies and corrects for directly. This study addresses these four gaps jointly: it applies an integrated predict-explain-locate framework to dense urban Canadian collision data, systematically compares balancing strategies, incorporates Canadian weather data, and audits its own feature set for label leakage rather than assuming engineered involvement features are safe to use at face value.

3 Data Description and Exploratory Data Analysis

3.1 Dataset Overview

Primary Dataset: Toronto Police Service Traffic Collisions Open Data (ASR-T-TBL-001), available through the Toronto Police Service Public Safety Data Portal [3]. This is the full collision dataset covering all severities (property damage, injury, and fatal), not the smaller Killed-or-Seriously-Injured (KSI) subset used in several prior studies. It contains 809,034 officially recorded traffic collision events between 2014 and 2026. Each record includes the collision date and time, GPS coordinates, neighbourhood identifier, vehicle and road-user involvement flags, injury outcome, and fatality count.

Secondary Dataset: Government of Canada Historical Climate Data, Toronto City Station (ID: 31688) [4]. Hourly weather observations including temperature (°C), relative humidity (%), dew point (°C), and precipitation amount (mm) were downloaded for all hours covered by the collision data. Wind speed and visibility fields were entirely null for this station and were excluded from the analysis.

The two datasets were merged on a date-hour key, achieving a 100% match rate across all 677,056 cleaned collision records meaning every collision timestamp fell within the weather station's hourly observation coverage, not that the matched weather condition is representative of every collision's specific location; as Section 8 notes, a single central station cannot capture hyperlocal weather variation across a geographically large city. After removing 131,978 records (16.3%) with invalid GPS coordinates (latitude/longitude = 0.0), the final working dataset contains 677,056 records, 44 columns after cleaning and feature engineering, and, after removal of the six leakage-affected columns identified in Section 3.4, **33 features** (21 categorical, 12 continuous) in the final model input matrix.

3.2 Descriptive Statistics

Table 1 presents descriptive statistics for the key numerical features in the final, corrected feature set. *Pedestrian Involved* and *Cyclist Involved* appeared in the original engineered feature set but were identified as leakage-affected (Section 3.4) and removed prior to modelling; they are omitted from this table accordingly.

Table 1: Descriptive statistics of key numerical features ($n = 677{,}056$).

Feature	Mean	Std Dev	Min	Max
Hour of Day (`OCC_HOUR`)	11.76	6.84	0	23
Month (`OCC_MONTH`)	6.48	3.42	1	12
Year (`OCC_YEAR`)	2019.8	3.21	2014	2026
Temperature (°C)	8.66	10.13	−21.8	28.1
Humidity (%)	70.31	16.42	18	100
Dew Point (°C)	3.11	12.18	−37.8	26.1
Precipitation (mm)	0.09	0.41	0.0	58.8
Is Rain Flag	0.058	0.234	0	1
Is Snow Flag	0.020	0.139	0	1
Is Ice Flag	0.012	0.110	0	1
Adverse Weather Flag	0.058	0.234	0	1
Is Weekend	0.236	0.425	0	1
Is Night (9 PM–5 AM)	0.127	0.333	0	1

3.3 Target Variable Distribution

Table 2 shows the three-class target variable distribution, revealing the severe class imbalance that is the central methodological challenge of this project.

Table 2: Target variable distribution showing severe class imbalance.

Class	Label	Record Count	Percentage
Minor (0)	Property damage only	580,492	85.7%
Major (1)	One or more injuries	95,928	14.2%
Fatal (2)	One or more fatalities	636	0.1%

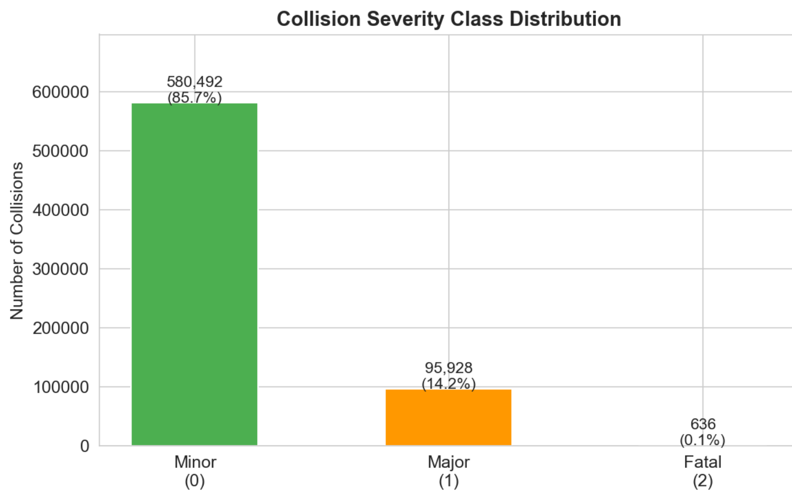

Figure 1: Collision Severity Class Distribution.

Figure 1 illustrates the severe class imbalance present in the dataset. A naive model predicting Minor for every collision would achieve 85.7% accuracy while completely failing to identify any dangerous outcome a benchmark this study returns to when evaluating whether headline accuracy is a meaningful metric in Section 5.

3.4 Data Preprocessing

The following preprocessing steps were applied in sequence during data cleaning and feature engineering.

1. **Timestamp conversion:** `OCC_DATE` was stored as a Unix millisecond timestamp. It was converted to UTC and then to Toronto local time (`America/Toronto`), floored to the nearest hour for weather merging.

2. **GPS missingness check (before dropping):** Prior to removing records with invalid GPS coordinates, the invalid-GPS rate was checked for association with collision severity, to confirm that dropping these records would not disproportionately remove records from the already-rare Fatal class. The check found the invalid-GPS rate varies substantially by severity 16.91% for Minor, 12.56% for Major, but only 4.50% for Fatal (a 12.4 percentage-point spread) meaning dropping invalid-GPS records is *not* class-neutral. Reassuringly, the direction favours the class of primary interest: Fatal collisions are the *least* likely to have invalid GPS (plausibly because fatal collisions receive more thorough on-scene investigation and location recording), so only 30 of 666 real Fatal records (4.50%) are lost to GPS cleaning, a smaller proportional loss than for Minor or Major.

3. **GPS cleaning:** 131,978 records (16.3%) with latitude/longitude = 0.0 (Not Spatially Available entries) were removed. All remaining records fall within Toronto's geographic boundaries.

4. **2026 partial-year check:** Because data collection extends into the current year, the number of months of 2026 data present was checked and flagged as a partial year, since `OCC_YEAR` is used as a model feature. Only 4 of 12 months are present for 2026 (15,467 records), a 19.3% shortfall against the same four months in 2025 (19,173 records) consistent with a partial year rather than a genuine year-over-year decline. Records were retained (individual collisions remain valid observations) but 2026 should not be compared directly against complete prior years in any trend analysis. *Note: the four months present for 2026 are January, February, March, and December not a contiguous year-to-date window, since `OCC_YEAR` is taken from the raw field while `OCC_MONTH` is separately re-derived from the timezone-converted `datetime` column.*

5. **Missing value imputation:** `FATALITIES` NaN was filled with 0 before creating the severity target variable. Involvement flags NaN values were filled with 'NO'. Numeric weather columns were filled with column medians.

6. **Data leakage audit and column removal:** Four involvement-flag columns `PEDESTRIAN`, `PASSENGER`, `BICYCLE`, and `MOTORCYCLE` were found to be populated in a manner correlated with collision severity (documented Fatal-rate ratios of 21.0× for pedestrian, 3.1× for passenger, 2.9× for bicycle, and 19.7× for motorcycle involvement, relative to the overall Fatal rate of 0.09%; see Section 4.2). These four raw columns, along with the two engineered features derived directly from them (`pedestrian_involved`, `cyclist_involved`), were removed prior to model training.

7. **Weather flag derivation:** Wind speed and visibility were entirely unavailable from the Toronto City station (100% null). Weather condition flags were derived numerically: `is_rain` (precipitation > 0.2 mm and temperature > 2°C), `is_snow` (precipitation $>$ 0.2 mm and temperature ≤ 2°C), `is_ice` (precipitation > 0.2 mm and temperature $\leq$ 0°C).

8. **Feature encoding:** Categorical columns (`OCC_DOW`, `season`, `hour_category`) were one-hot encoded. The neighbourhood column (158 unique values) was frequency-encoded to a single continuous variable representing collision frequency per neighbourhood.

9. **Severity target:** The target variable was derived as a three-class ordinal variable: Minor (0) if `FATALITIES` = 0 and `INJURY_COLLISIONS` = 'NO'; Major (1) if `INJURY_COLLISIONS` = 'YES' and `FATALITIES` = 0; Fatal (2) if `FATALITIES` ≥ 1.

3.5 Feature Engineering

The following features were engineered from the raw data to improve model performance. *`pedestrian_involved` and `cyclist_involved` are listed here for completeness with the original pipeline but were removed prior to modelling, per Section 3.4.*

- `season`: Winter / Spring / Summer / Fall derived from month number.
- `hour_category`: Peak (7–9 AM and 4–7 PM weekdays), Offpeak, Late Night (9 PM–5 AM), Weekend.
- `is_weekend`: Binary flag (1 = Saturday or Sunday).
- `is_night`: Binary flag (1 = 9 PM to 5 AM).
- `pedestrian_involved` *(removed leakage)*: originally a binary flag derived from the `PEDESTRIAN` involvement column.
- `cyclist_involved` *(removed leakage)*: originally a binary flag derived from the `BICYCLE` involvement column.
- `neighbourhood_freq`: Frequency encoding of the 158 Toronto neighbourhood categories into a single continuous collision-frequency score.
- Weather flags: `is_rain`, `is_snow`, `is_ice` derived numerically from precipitation and temperature.

The final feature matrix contains 33 features (21 categorical, 12 continuous) across all 677,056 cleaned records (Section 3.1).

3.6 Exploratory Data Analysis

A set of exploratory visualizations was produced during the exploratory data analysis stage to examine temporal, spatial, and environmental patterns. None of these visualizations reference the removed leakage columns, so they remain valid without regenerating them; they are reported here as originally produced.

3.6.1 Temporal Patterns: Hour of Day

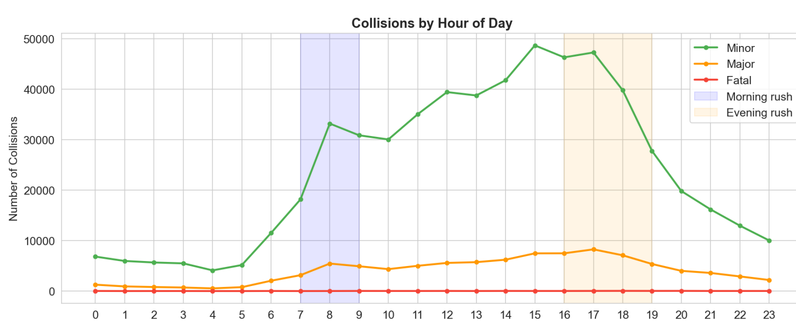

Figure 2: Collisions by Hour of Day, by Severity Class.

Figure 2 shows that Minor collisions follow a clear rush-hour pattern, peaking sharply during morning (7–9 AM) and evening (4–7 PM) rush hours. Late-night hours (10 PM to 4 AM) show lower absolute collision counts but a proportionally higher rate of Major and Fatal outcomes. Possible explanations include differences in vehicle speed, traffic exposure, driver fatigue, and impaired driving; however, none of these mechanisms were directly measured in the present dataset, so this pattern is reported as an observed association rather than an established cause. This temporal pattern directly justifies the creation of the `hour_category` and `is_night` engineered features.

3.6.2 Temporal Patterns: Monthly and Weekly

The supplementary monthly and day-of-week plots are provided in Appendix A. They show clear temporal variation in collision volume, including strong seasonal variation and a high Friday collision count, supporting the use of `season` and day-of-week features in the modelling pipeline. Per the partial-year finding (Section 3.4), 2026 is under-represented in any monthly breakdown that includes it (4 of 12 months present) and should not be read as showing an actual decline in collision volume for that year.

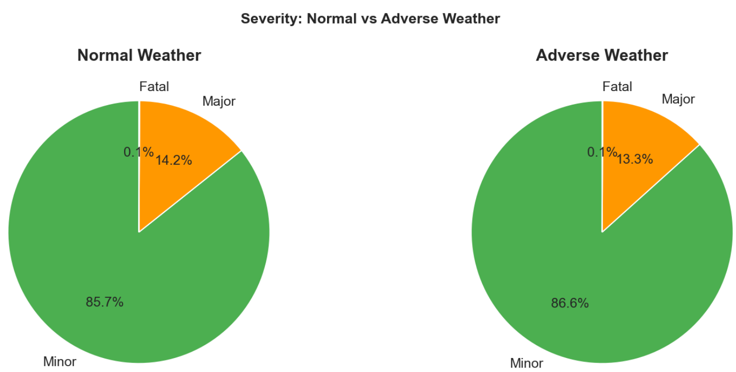

Figure 3: Severity Distribution: Normal vs. Adverse Weather.

3.6.3 Weather Effects on Severity

Figure 3 reveals a counterintuitive pattern: under adverse weather (rain, snow, or ice), the proportion of Major collisions decreases slightly. One possible explanation is the **risk compensation effect** described in traffic safety research when conditions are clearly dangerous, drivers may consciously reduce speed and increase following distance, reducing the severity of collisions that do occur even as total collision volume increases although this mechanism was not directly tested in the present study and the pattern should be read as an observed association. This non-linear relationship between weather and severity further motivates the use of non-linear ensemble models over a purely linear baseline.

3.6.4 Feature Correlation Analysis

The supplementary correlation heatmap in Appendix A shows a strong correlation between temperature and dew point (approximately 0.95), while direct linear correlations between individual predictors and severity are small. This supports treating the problem as potentially non-linear and motivates the comparison of flexible ensemble models against a linear baseline.

3.6.5 Spatial and Neighbourhood Patterns

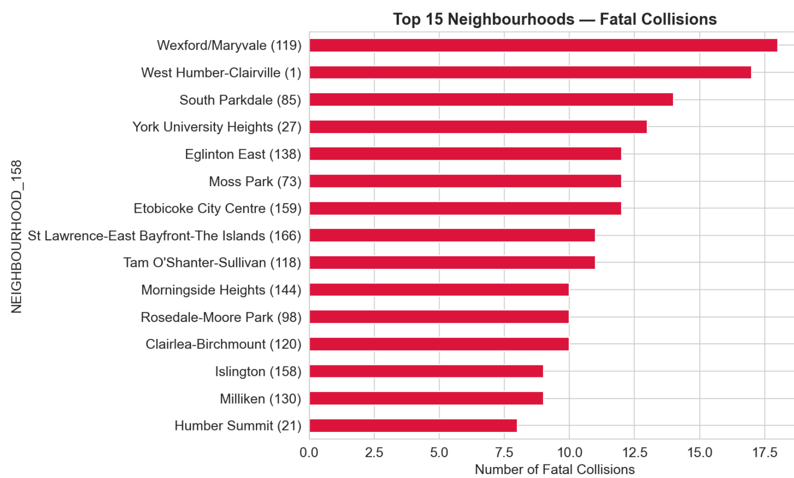

Figure 4: Top 15 Toronto Neighbourhoods by Fatal Collision Count (2014–2026).

Figure 4 identifies that Wexford/Maryvale leads with 18 fatal collisions across the study period. These top neighbourhoods contain major arterial corridors and warrant further investigation of factors such as speed, pedestrian exposure, cycling exposure, and crossing infrastructure none of which were systematically measured in this dataset, so this observation is reported as a research implication rather than an established cause. This finding directly informed the inclusion of the frequency-encoded neighbourhood variable. Its full corrected SHAP ranking is reported in Section 5.6: `neighbourhood_freq` and `HOOD_158` (the frequency-encoded and raw neighbourhood identifiers, respectively) rank fourth and fifth globally in the corrected model, and `HOOD_158` is the fourth-strongest feature in the Fatal-specific ranking both location features are consistently among the model's top predictors, alongside `is_weekend` and time-of-year features.

4 Methodology

4.1 Research Design Overview

This research applies a supervised machine learning methodology, integrating data from two publicly available sources, auditing the feature set for label leakage, addressing class imbalance through a categorical-aware oversampling strategy compared against alternatives, training and comparing five classification algorithms via cross-validation, and applying SHAP explainability and DBSCAN spatial clustering to the corrected model.

Data split: The cleaned records were divided into stratified subsets: a training set (70%) used for model learning and SMOTE-NC resampling; a validation set (15%) used for *every* selection decision reported in Section 5 the single-split model comparison, the balancing-strategy comparison, threshold tuning, and the tuned-vs-default hyperparameter comparison; and a held-out test set (15%), reserved untouched for the explainability analysis (Section 5.6) and not used for any selection decision in this study. This validation/test separation is itself a methodological correction: an earlier version of this pipeline created a validation split but did not use it, instead reusing the same test split for both selection decisions and final reporting, which risked the selected model/strategy/threshold appearing more favourable on that same split than it would on genuinely unseen data. Stratified sampling (`stratify=y, random_state=42`) was applied to ensure each split preserved the original class proportions. The primary reported model comparison (Section 5.1), however, uses full-dataset cross-validation rather than this single split, per Section 4.6.

What each data partition was used for, and a disclosed limit on their independence. To state this precisely: the training split fits each model; the validation split selects the balancing strategy (Section 4.2, using XGBoost as a fixed base classifier), the classification threshold (Section 5.3), and compares tuned against default hyperparameters (Section 5.4); the test split is reserved solely for the SHAP explainability analysis (Section 5.6) and enters no selection decision. The headline cross-validated model comparison (Table 3, Section 5.1), however, is computed on the *full* 677,056-record dataset that is, the training, validation, and test records pooled and re-partitioned into five fresh CV folds, independent of the fixed train/validation/test split used for selection. This means the headline CV figures are not a strictly independent, nested-CV generalization estimate relative to the validation-split-based strategy and threshold decisions: the same records that informed the choice of SMOTE-NC as the balancing strategy also appear, redistributed into different folds, in Table 3. This is disclosed here explicitly rather than presented as a fully independent estimate; a fully nested cross-validation, in which strategy selection is repeated inside each outer fold, was not performed given the computational cost of refitting SMOTE-NC and five classifiers per fold repeatedly. The magnitude of any resulting optimism was not quantified and remains a real limitation of the current design, treated as such in Section 8, not minimized.

4.2 Handling Class Imbalance

With Fatal collisions representing only approximately 0.1% of records, a naive classifier predicting Minor for every record achieves 85.7% accuracy while detecting no fatal outcomes.

SMOTE-NC. SMOTE (Synthetic Minority Oversampling Technique) generates new synthetic minority-class samples by interpolating between existing examples in feature space. However, standard SMOTE treats every feature as continuous, including the one-hot encoded categorical columns produced during feature engineering (e.g. `season_Spring`, `is_weekend`). Interpolating between binary indicator columns produces fractional values (e.g. 0.37) that correspond to no real collision. This project therefore uses **SMOTE-NC** (`SMOTENC` from the `imbalanced-learn` library), which correctly treats categorical columns as categorical sampling them by nearest-neighbour majority vote rather than interpolation while still interpolating genuinely continuous features (temperature, humidity, GPS coordinates, neighbourhood frequency). SMOTE-NC was applied exclusively to the training set; validation and test sets retained the original imbalanced distribution to reflect real-world conditions.

One-hot group encoding correction. A second, methodologically distinct issue arises once the day-of-week, season, and hour-category variables have already been one-hot encoded into separate binary columns (Section 3.4). Supplying these already-split columns to `SMOTENC` causes each column within a group to be resampled independently, by its own nearest-neighbour majority vote, with no constraint linking columns that belong to the same original categorical variable: a synthetic row can end up with zero, or in principle more than one, of the seven day-of-week columns set to 1, violating the mutual exclusivity that one-hot encoding is meant to guarantee. This was not a theoretical concern: across the resampled training data, 34.8% of day-of-week rows, 15.0% of hour-category rows, and 5.8% of season rows did not sum to exactly 1 within their one-hot group. The correction collapses each one-hot group back into a single categorical column immediately before SMOTE-NC runs, so that day-of-week, for example, is sampled as one seven-level categorical feature rather than seven independent binary features, and re-expands the result to one-hot columns immediately afterward. Two checks validate this procedure: a round-trip check confirming that collapsing and re-expanding exactly reproduces the original one-hot columns, and a post-resampling check confirming that every one-hot group sums to exactly 1 per row; both passed with zero violations across all three groups. The corrected resampling produced 406,344 training records per class (1,219,032 total), unchanged from the uncorrected total, since the correction affects which categories are assigned rather than how many synthetic rows are generated.

Data leakage audit. Before finalizing the balancing strategy, the involvement-flag columns `PEDESTRIAN`, `PASSENGER`, `BICYCLE`, and `MOTORCYCLE` were audited for label leakage by comparing the Fatal-class rate among records where each flag was set against the overall Fatal rate (0.09%). This audit found Fatal rates 21.0× higher when `PEDESTRIAN` was set, 19.7× higher for `MOTORCYCLE`, 3.1× higher for `PASSENGER`, and 2.9× higher for `BICYCLE`, relative to the dataset-wide Fatal rate consistent with these fields being populated, at least in part, as a consequence of injury/fatality outcome recording rather than as independently observable pre-collision predictors. All four raw columns, and the two engineered features derived from them, were removed prior to final model training (Section 3.4).

General leakage rule and a final audit of the remaining feature set. The four involvement-flag columns were identified because their SHAP importance in the original, uncorrected model was implausibly high for fields that should, in principle, be recorded independently of collision outcome; the general rule applied here, and recommended for any similar dataset, is that *no predictor should encode information that would only become known after injury severity or the collision outcome had been established.* Before finalizing the feature set, the remaining 33 features were reviewed against this rule by category rather than assumed safe by default. Time features (`OCC_HOUR`, `OCC_MONTH`, `OCC_YEAR`, `is_weekend`, `is_night`, `hour_category`, day-of-week) are recorded at the moment of collision and cannot be influenced by its eventual severity. Weather features (temperature, humidity, dew point, precipitation, `is_rain`/`is_snow`/`is_ice`) are drawn from an independent government climate dataset merged on date-hour and are similarly outcome-independent. Location features (GPS coordinates, `HOOD_158`, `neighbourhood_freq`) describe where a collision occurred, not how severe it was. No feature in the final set was found to plausibly encode post-outcome information under this review; as stated in Section 8, this audit cannot rule out subtler, undetected leakage that produces a less conspicuous SHAP signal than the four removed flags did.

Five balancing strategies were compared using XGBoost as the base classifier:

1. **No Synthetic Resampling (XGBoost weighting configuration):** no synthetic samples were generated. The experiment used the implementation-specific `scale_pos_weight=3` setting present in Notebook 4. Because this parameter is not a general multiclass class-weighting mechanism, this branch is treated only as a no-synthetic-resampling comparison condition and not as a definitive test of optimal three-class weighting.

2. **SMOTE-NC:** oversamples Minor, Major, and Fatal to equal class sizes, correctly handling categorical features (primary strategy).

3. **ADASYN:** similar to SMOTE but generates proportionally more samples in harder-to-learn decision boundary regions. *Note: ADASYN does not have a categorical-aware variant in the `imbalanced-learn` library and therefore retains the same categorical-interpolation limitation as plain SMOTE.*

4. **SMOTETomek:** combines SMOTE oversampling with Tomek Links cleaning to remove borderline noisy majority-class samples. *Carries the same categorical-interpolation caveat as ADASYN.*

5. **Random Undersampling:** reduces the majority class (Minor) by randomly removing records until all three classes are equal.

4.3 Model Selection and Configuration

Five classification models were selected to span a range of algorithmic complexity, from a simple linear baseline to advanced ensemble and deep learning methods.

- **Logistic Regression (Baseline):** Linear classifier using the `saga` solver (`max_iter=500, class_weight='balanced'`). Included as a linear baseline following Al-Ghamdi [1].
- **Random Forest:** 100 independent decision trees (`n_estimators=100, max_depth=10, class_weight='balanced'`). Included following Chen and Wang [2].
- **XGBoost (Primary Model):** 200 sequential gradient-boosted trees (`n_estimators=200, max_depth=6, learning_rate=0.1`). Selected following Parsa et al. [6], who identified XGBoost as the top performer for collision severity prediction.
- **Gradient Boosting Machines:** 100 sequential trees (`n_estimators=100, max_depth=4, learning_rate=0.1, subsample=0.8`). Included following Sheykhfard et al. [9].
- **Neural Network (MLP):** Two hidden layers with 64 and 32 units, early stopping to prevent overfitting on the SMOTE-NC-resampled training set. Included following Sameen and Pradhan [8].

XGBoost is retained as this study's primary model for the SHAP explainability analysis (Section 4.7) on the strength of its F1 Macro and Accuracy lead among the five classifiers (Section 5.1). This designation does not extend to Fatal-class discrimination specifically: Random Forest achieves the highest Fatal-class PR-AUC of the five models (Section 5.1), so the two models are best understood as leading on different metrics rather than one being an unqualified best performer.

4.4 Evaluation Metrics

The following metrics were computed for each model:

- **Accuracy:** proportion of correctly classified records overall. *Reported for completeness only with an 85.7% majority-class baseline, accuracy alone is close to meaningless for this problem and is not used to rank models.*
- **F1 Weighted:** F1 score weighted by class support.
- **F1 Macro (primary ranking metric):** F1 score averaged equally across all three classes, penalizing poor Fatal-class performance.

- **ROC-AUC (One-vs-Rest, Weighted):** class separation ability across decision thresholds.
- **PR-AUC (Fatal class):** Precision-Recall AUC computed for the Fatal class specifically. At approximately 0.1% Fatal prevalence, ROC-AUC can appear favourable even for a model with little genuine ability to detect Fatal collisions, since it is dominated by performance on the much larger Minor class; PR-AUC is a more informative metric for this degree of imbalance.

4.5 Hyperparameter Tuning

This subsection describes an **exploratory hyperparameter search**, not a reliably tuned model: `RandomizedSearchCV` with 20 random parameter combinations and 3-fold stratified cross-validation was applied to XGBoost and Random Forest, optimizing for F1 Macro on the SMOTE-NC-resampled training set, and final parameter adoption was judged on the held-out validation set (Section 5.4), not on the search's own reported score. *Methodological note: this internal cross-validation is performed on data that has already been resampled, meaning synthetic points in one fold may be derived from real or synthetic points that appear in another fold. This is a narrower-scope version of the same resampling-before-splitting leakage risk that motivated the corrected full-dataset cross-validation described in Section 4.6, and should be interpreted with that caveat the internal cross-validated score reported by the search process is expected to be optimistic relative to genuine held-out performance. This is not a hypothetical concern: the search reported an internal best F1 Macro of 0.9268 for XGBoost, yet the resulting tuned model performs worse than the untuned default when evaluated on the real, unresampled validation split (Section 5.4) a direct, measured illustration of exactly the inflation this caveat warns about, not merely a theoretical risk.* **The 0.9268 (XGBoost) and 0.7884 (Random Forest) internal search scores reported in Section 5.4 are not directly comparable to the validation-split figures in Table 6 or the cross-validated figures in Table 3**, and should not be read as this study's actual model performance. The Random Forest search was conducted on a 200,000-record sample for computational feasibility, drawn with a seeded random-state generator (`np.random.RandomState(RANDOM_STATE)`) so this subsample is exactly reproducible.

4.6 Cross-Validation

Five-fold stratified cross-validation was applied to all five models on the *full* cleaned dataset (not a subsample), with SMOTE-NC resampling performed *inside* each fold via an `imblearn` pipeline rather than once before the cross-validation loop. This avoids a subtler form of leakage in which synthetic training points generated from data outside a given fold could influence that fold's evaluation. Accuracy, F1 Weighted, F1 Macro, ROC-AUC, and PR-AUC (Fatal) were computed per fold; mean and standard deviation across folds are reported as the primary model comparison (Section 5.1).

4.7 SHAP Explainability

SHAP `TreeExplainer` was applied to the final XGBoost model on a sample from the test set. *Important caveat: the explained model was trained on SMOTE-NC-resampled data, i.e. a mix of real and synthetic collision records. SHAP attributions therefore describe what the model learned from this resampled training distribution, not a causal or purely empirical relationship observed directly in the real collision data. This is stated explicitly here so that SHAP results in Section 5.6 are not over-interpreted as direct empirical findings.* SHAP values were computed with one value per record, feature, and class, and reshaped into class-specific sub-arrays. Four visualizations were produced: global stacked feature importance across all classes; a beeswarm plot for Fatal predictions; a beeswarm plot for Major predictions; and a waterfall plot for a single fatal collision case.

4.8 Geospatial Analysis

DBSCAN spatial clustering was applied to the GPS coordinates of all confirmed fatal collision records. Parameters: `eps` = 300/6,371,000 (300 metres converted to radians), `min_samples` = 3, `metric` = 'haversine'. This component of the analysis uses only GPS coordinates and the (unaffected) severity label, and is therefore not impacted by the leakage correction or the SMOTE-NC fix. Three interactive Folium HTML maps were generated: a full heatmap of all cleaned collision records, a fatal-collision overlay map, and a DBSCAN cluster-zone visualization.

5 Results

5.1 Experiment 1: Model Performance Comparison

Table 3 reports the corrected, headline model comparison: 5-fold stratified cross-validation on the full 677,056-record dataset, with SMOTE-NC applied inside each fold, for all five models. Values are mean $\pm$ standard deviation across folds.

Table 3: Cross-validated model performance, full dataset, 5-fold stratified CV, mean ± SD. Sorted by F1 Macro (primary ranking metric). The Dummy baseline (bottom row) is a single validation-split evaluation, not a 5-fold CV figure, and is included for reference rather than as a like-for-like comparison row.

Model	Accuracy	F1 Weighted	F1 Macro	ROC-AUC	PR-AUC (Fatal)
XGBoost	0.8142 ± 0.0043	0.7851 ± 0.0016	$\mathbf{0.3381 \pm 0.0014}$	0.5465 ± 0.0027	0.0012 ± 0.0002
Gradient Boosting	0.6688 ± 0.0098	0.7106 ± 0.0063	0.3338 ± 0.0020	0.5355 ± 0.0021	0.0014 ± 0.0003
Random Forest	0.5501 ± 0.0061	0.6314 ± 0.0041	0.3057 ± 0.0018	0.5229 ± 0.0011	$\mathbf{0.0017 \pm 0.0004}$
Neural Network	0.4965 ± 0.0352	0.5811 ± 0.0303	0.2871 ± 0.0110	0.5135 ± 0.0051	0.0011 ± 0.0001
Logistic Regression	0.4280 ± 0.0101	0.5329 ± 0.0092	0.2668 ± 0.0034	0.5152 ± 0.0026	0.0010 ± 0.0001
Dummy (Majority Class)†	0.8574	0.7915	0.3077		0.0009

†Single validation-split evaluation (not 5-fold CV); ROC-AUC not computed, since a constant-prediction classifier has no probability output to rank on.

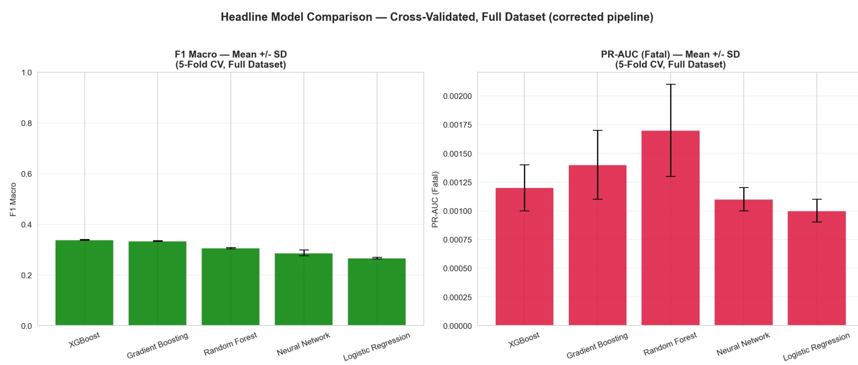

Figure 5: Headline Model Comparison: F1 Macro and PR-AUC (Fatal).

Three findings stand out. First, Fatal-class PR-AUC is low in absolute terms across all five model families, ranging from 0.0010 to 0.0017. For reference, a no-skill Precision–Recall AUC is approximately equal to the positive-class prevalence; with Fatal prevalence near 0.09%, the corresponding reference level is approximately 0.0009. These values are not claimed to be statistically indistinguishable from the reference because no formal test of that difference was performed; they are instead described as practically weak Fatal-class discrimination. Second, the metric hierarchy produces two different leaders. XGBoost has the highest F1 Macro (0.338) and Accuracy, making it the strongest overall three-class model under the primary aggregate criterion, whereas Random Forest has the highest Fatal-class PR-AUC (0.0017), making it the strongest of the evaluated models on the Fatal-specific discrimination metric. Neither model is therefore an unqualified overall winner. Third, the broadly weak Fatal-specific results across several model families support the interpretation that the current predictor set provides limited Fatal-class separability under the present modelling setup. This interpretation is supported by the consistency of the pattern across different model families, but it does not rule out limitations of the current modelling choices or the possibility that richer predictors, alternative model classes, or different evaluation designs could perform better.

5.1.1 Majority-Class Baseline

A trivial classifier that always predicts the majority class (Minor) was evaluated on the validation split as a contextual diagnostic reference. It achieves Accuracy 0.8574, F1 Weighted 0.7915, F1 Macro 0.3077, and Fatal PR-AUC 0.0009. The high accuracy illustrates why accuracy is not sufficient for this highly imbalanced task. For the Fatal-class Precision–Recall metric, the more general no-skill reference is approximately the Fatal prevalence itself (about 0.0009). The five cross-validated model PR-AUC values of 0.0010–0.0017 are therefore close in absolute terms to this reference level and are interpreted as practically weak, without making a formal claim that the differences are statistically indistinguishable.

5.2 Experiment 2: Balancing Strategy Comparison

Table 4 compares five balancing strategies using XGBoost as the base classifier, evaluated on the validation split (Section 4.1) so that the strategy selected here is not chosen using the test split reserved solely for the SHAP explainability analysis.

Table 4: Balancing strategy comparison using XGBoost as the base classifier, validation split.

Strategy	Train Size	Accuracy	F1 Macro	Recall Fatal	PR-AUC (Fatal)
SMOTE-NC	1,219,032	0.8102	**0.3370**	0.0000	0.0011
ADASYN	1,229,713	0.8564	0.3109	0.0000	0.0012
SMOTETomek	1,204,024	0.8562	0.3106	0.0000	0.0012
No Synthetic Resampling	473,939	0.8574	0.3082	0.0000	0.0018
Random Undersampling	1,335	0.3733	0.2456	**0.4105**	0.0014

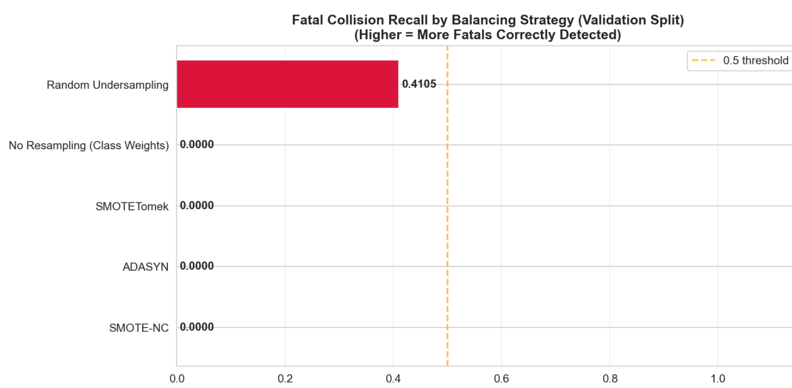

Figure 6: Fatal Collision Recall by Balancing Strategy.

Four of the five configurations SMOTE-NC, ADASYN, SMOTETomek, and No Synthetic Resampling achieve 0.0000 Fatal recall on the validation split at their default decision rules. Random Undersampling achieves the highest Fatal recall (41.05%), but overall accuracy falls to 37.33% and F1 Macro to 0.2456 after the training data are reduced to 1,335 observations. SMOTE-NC achieves the highest F1 Macro (0.3370) among the tested balancing configurations, while the no-synthetic-resampling branch has the highest validation Fatal PR-AUC (0.0018) in this particular comparison. Corrected SMOTE-NC produces exactly equal class sizes (406,344 observations per class), so its weak Fatal recall cannot be explained by continued Minor-class numerical dominance after resampling. Rather, the experiment shows that equalizing class counts does not by itself create strong class separability; synthetic oversampling increases minority representation but cannot guarantee that the available predictors contain sufficiently distinctive information for Fatal observations. No tested strategy simultaneously achieves strong overall three-class performance and strong Fatal-class detection.

5.3 Experiment 3: Probability Threshold Tuning

Table 5 reports Fatal recall, precision, and overall accuracy at six classification thresholds for the corrected XGBoost model, evaluated on the validation split.

Table 5: Threshold tuning results for the Fatal class, corrected XGBoost model, validation split.

Threshold	Fatal Recall	Fatal Precision	Overall Accuracy
0.05	**0.5895**	0.0011	0.4297
0.10	0.3579	0.0013	0.6036
0.15	0.1789	0.0011	0.6966
0.20	0.0632	0.0007	0.7477
0.30	0.0211	0.0008	0.7952
0.50	0.0000	0.0000	0.8145

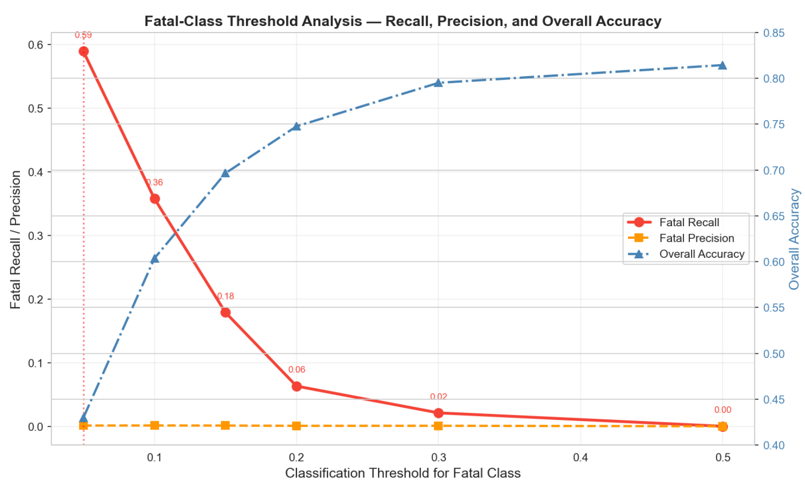

Figure 7: Fatal-Class Threshold Analysis: Recall, Precision, and Overall Accuracy.

Among the evaluated thresholds, 0.05 produces the highest Fatal recall, increasing recall from 0.0% at 0.50 to 58.95%. However, overall accuracy decreases from 81.45% to

42.97%, and Fatal precision at 0.05 is only 0.11%. The experiment therefore demonstrates a substantial recall–false-positive trade-off. Lowering the threshold changes the decision rule rather than the information learned by the model, so the recall increase should not be interpreted as evidence of stronger underlying predictive signal. The 0.05 operating point is reported only as the highest-recall threshold among those tested and is not recommended for operational deployment.

5.4 Experiment 4: Hyperparameter Tuning

`RandomizedSearchCV` was applied to XGBoost and Random Forest on the SMOTE-NC-resampled training set (Section 4.5). Table 6 compares default and tuned configurations on the validation split. Recall the caveat from Section 4.5: the search's own internal cross-validated score is optimistic, since it is computed on already-resampled data the search reported an internal best F1 Macro of **0.9268** for XGBoost and **0.7884** for Random Forest; the values below are the tuned models' performance on the real, unresampled validation split, which does not carry that caveat, and the gap between these two sets of numbers is itself the finding.

Table 6: Default versus tuned model performance, validation split.

[1]

Model	Accuracy	F1 Weighted	F1 Macro	F1 Fatal	Recall Fatal
Random Forest (Tuned)	0.6793	0.7168	**0.3393**	0.0012	0.0105
XGBoost (Default)	0.8102	0.7834	0.3370	0.0000	0.0000
XGBoost (Tuned)	0.8518	0.7935	0.3182	0.0000	0.0000
Random Forest (Default)	0.5574	0.6375	0.3077	0.0024	0.1789

The corrected results show that tuning is, at best, a mixed intervention and, in XGBoost's case, actively counterproductive on real held-out data. XGBoost's F1 Macro *decreases* under tuning (0.3370 $\rightarrow$ 0.3182), and its Fatal recall stays at exactly zero either way. Random Forest's F1 Macro improves under tuning (0.3077 $\rightarrow$ 0.3393), the largest single change in this table, but this comes with a striking hidden cost: Random Forest's Fatal recall *collapses* from 17.89% (default) to 1.05% (tuned) the tuned model looks better on the metric the search optimized for (F1 Macro) while becoming dramatically worse at the one outcome this study cares about most. Neither model's internal search score (0.9268 XGBoost, 0.7884 Random Forest) bears any resemblance to its real validation performance (F1 Macro 0.318–0.339 for either), which is direct, measured confirmation of the internal-CV leakage caveat raised in Section 4.5: `RandomizedSearchCV`'s internal cross-validation, run on SMOTE-NC-resampled folds, produces scores that should never be reported as real model performance. **The practical conclusion is that hyperparameter tuning, as configured here, should not**

[1]The Random Forest hyperparameter search subsamples 200,000 training rows, drawn via a seeded `np.random.RandomState(RANDOM_STATE)` generator in the model-training pipeline. The figures reported above are from a single, exactly reproducible run.

be trusted to improve and in at least one case actively harms Fatal-class detection, and any future use of this pipeline should evaluate tuned models against held-out, unresampled data before adopting tuned parameters, exactly as done here.

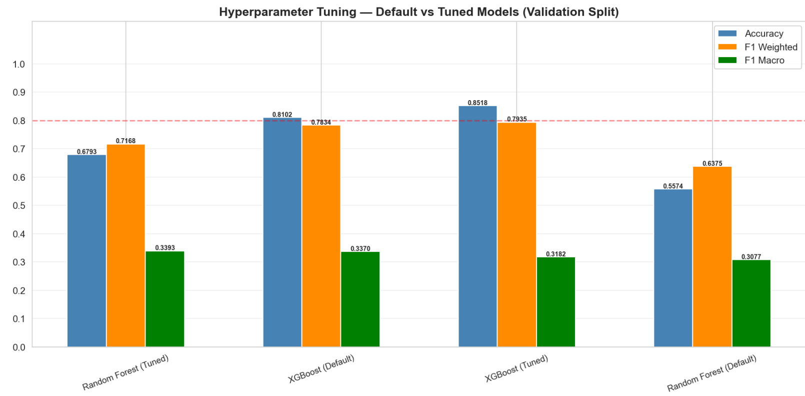

Figure 8: Hyperparameter Tuning: Default vs. Tuned Model Performance.

5.5 Experiment 5: Cross-Validation

The primary analysis therefore uses a single, unified full-dataset, all-model, 5-fold cross-validation framework (Section 4.6), reported directly in Table 3 (Section 5.1), rather than a two-experiment structure combining a single-split comparison across all models with a separate subsample-based cross-validation of one model an earlier design whose figures were difficult to reconcile with one another.

The standard deviations in Table 3 are informative in their own right. The best-performing model by F1 Macro, XGBoost, has a fold-to-fold standard deviation of 0.001 on F1 Macro and 0.0002 on PR-AUC (Fatal). The fold-to-fold variation is small, indicating **stable** performance across the five folds (though not uniquely so on every metric Logistic Regression and the Neural Network post smaller PR-AUC standard deviations, at 0.0001 each). This stability, however, demonstrates only that the model behaves consistently across different random partitions of the data; it does not by itself demonstrate that the model **generalizes** to genuinely new, future collisions, nor does it change the substantive finding that the stable value being reproduced across folds is itself close to the no-skill floor for Fatal detection. A model can be highly stable and simultaneously uninformative for the class of primary interest, which is the situation observed here.

5.6 SHAP Explainability Results

Caveats (Section 4.7). Two apply here, and the second is the more consequential one. First, the SHAP values reported here explain XGBoost as trained on SMOTE-NC-resampled data a mix of real and synthetic collision records not a relationship measured directly on real, unresampled collisions. Second, and more importantly given Section 5.1's finding that XGBoost's Fatal-class PR-AUC (0.0012) is close to the no-skill baseline (0.0009): SHAP explains what a fitted model relies on to produce its predictions, and it does this regardless of whether those predictions are any good. Because the underlying model has very limited genuine discriminative ability for the Fatal class, the ranking below should be read as a description of the model's internal reasoning useful for auditing what the model is doing and for confirming the one-hot fix's effect (below) and not as evidence that the ranked features are reliable, real-world Fatal-collision risk factors. Explanations of a weak predictive model are themselves of limited substantive value, independent of whether the explanation method is applied correctly; this caveat is more important than the resampled-data caveat for interpreting what follows. XGBoost is retained as the explained model both as this study's stated primary model (Section 4.3) and as the top performer by F1 Macro in the corrected cross-validated comparison (Section 5.1).

This section reflects a full rerun against the corrected model, after the one-hot/SMOTE-NC encoding fix (Section 4.2) was applied and the underlying model retrained. The ranking below is substantially different from the pre-fix ranking, and this difference is itself a finding: the pre-fix SHAP explanations were generated against a model trained on one-hot columns in which 5.8–34.8% of rows had corrupted group-level values (Section 4.2), so a SHAP ranking dominated by individual day-of-week indicators is consistent with SHAP having partly picked up on that corruption artifact rather than a genuine day-of-week effect. With the corruption fixed, individual day-of-week columns barely register in the corrected ranking.

Table 7: Global SHAP feature importance ranking (all classes combined), corrected model (XGBoost, trained on SMOTE-NC-resampled data), top 15 by total stacked mean |SHAP|.

Rank	Feature
1	`OCC_YEAR`
2	`is_weekend`
3	`OCC_MONTH`
4	`neighbourhood_freq`
5	`HOOD_158`
6	`humidity`
7	`is_night`
8	`OCC_HOUR`
9	`hour_category_offpeak`
10	`LAT_WGS84`
11	`temperature`
12	`season_Spring`
13	`hour_category_weekend`
14	`is_rain`
15	`season_Winter`

Two changes from the pre-fix ranking are immediately apparent. First, `PEDESTRIAN` the single strongest directional predictor in the original, leakage-affected model, before that column was removed (Section 3.4) remains absent, confirming the leakage removal continues to hold under the corrected encoding. Second, and more notable: **individual day-of-week indicators, which previously occupied six of the top ten Fatal-specific ranking positions, are now essentially absent from the top 15** (only `OCC_DOW_Monday` appears at all, and only at rank 15 of the Fatal-specific beeswarm below). In their place, the corrected ranking is led by `OCC_YEAR`, `is_weekend` (the day-level binary flag, rather than individual weekday dummies), `OCC_MONTH`, and location/neighbourhood features a ranking with a more conventional, defensible structure than the fragmented, day-of-week-heavy pre-fix result. `OCC_YEAR`'s top rank should still be read alongside the 2026 partial-year caveat (Section 3.4): with only 4 of 12 months present for the most recent year, any year-level pattern in this ranking should not be over-interpreted as a clean secular trend.

The Fatal-class-specific SHAP beeswarm (Figure 10) gives the ranking specific to Fatal predictions: `is_weekend`, `OCC_YEAR`, `OCC_MONTH`, `HOOD_158`, `humidity`, `hour_category_weekend`, `is_night`, `hour_category_offpeak`, `is_rain`, and `temperature` (top 10). Directionally, `is_weekend` = 1 pushes the Fatal-class prediction strongly negative (away from Fatal), while `is_weekend` = 0 (weekday collisions) clusters closer to neutral or slightly positive in this corrected model, weekday collisions carry more of the Fatal-class signal than weekend ones, the reverse of what a purely exposure-based story (more driving on weekends) might predict, and worth flagging as a finding for further investigation rather than an obvious result. `is_rain` = 1 also pushes strongly negative, consistent with the risk-compensation pattern already observed in the EDA (Section 3.6.3). `hour_category_weekend` (a hour-of-day bucket, distinct from the day-level `is_weekend` flag) shows the opposite pattern, pushing clearly positive when active the two features capture different aspects of "weekend-ness" and are not contradictory.

This pattern should still be interpreted alongside Section 5.1's finding that the corrected model's Fatal-class PR-AUC is only 0.0010–0.0017, close to the no-skill floor. Even with the encoding artifact resolved, a model with this little genuine signal will still attribute importance to *something*, and with a real Fatal sample of only 636 collisions (before SMOTE-NC oversampling), this ranking should be read as the best available explanation of what the model is doing, not as a validated set of causal risk factors. Compared to the pre-fix ranking, however, this is a materially more defensible result, and this study treats the improvement itself a shift from an implausible, artifact-driven ranking to a conventional, exposure/location/weather-driven one as evidence that the one-hot fix (Section 4.2) was not merely a technical correctness issue but one that materially affected the substantive interpretability of the model.

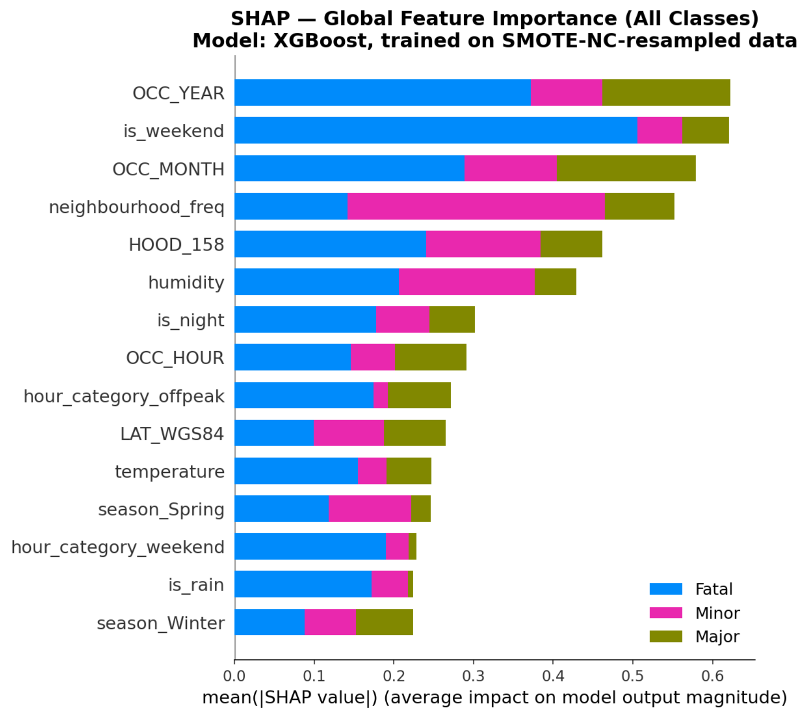

Figure 9: SHAP Global Feature Importance.

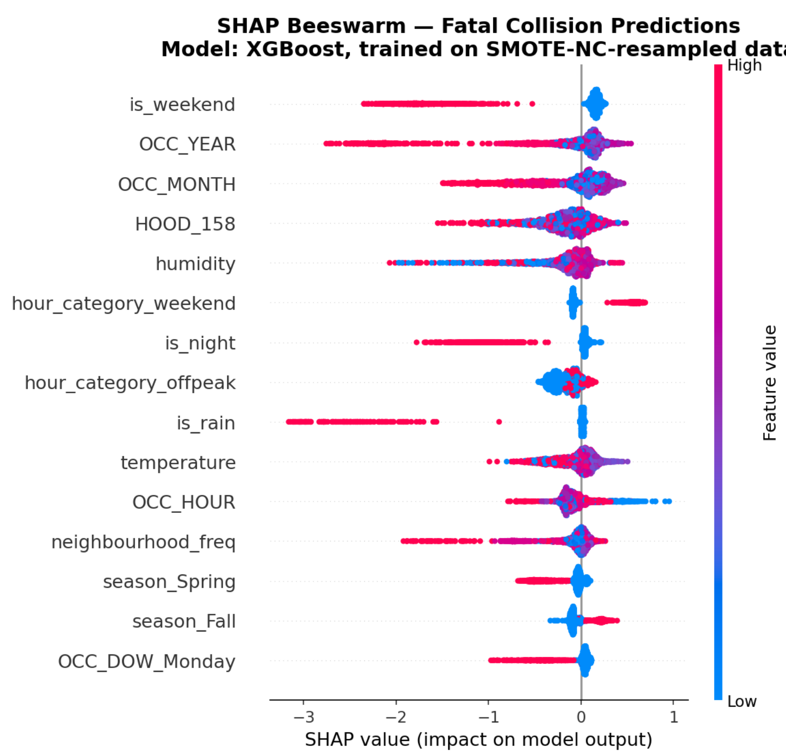

Figure 10: SHAP Beeswarm, Fatal Class.

5.7 Geospatial Analysis Results

This subsection is unaffected by the leakage and SMOTE-NC corrections, since DBSCAN operates only on GPS coordinates and the (unchanged) severity label. Values are retained from the original run and are not expected to change materially.

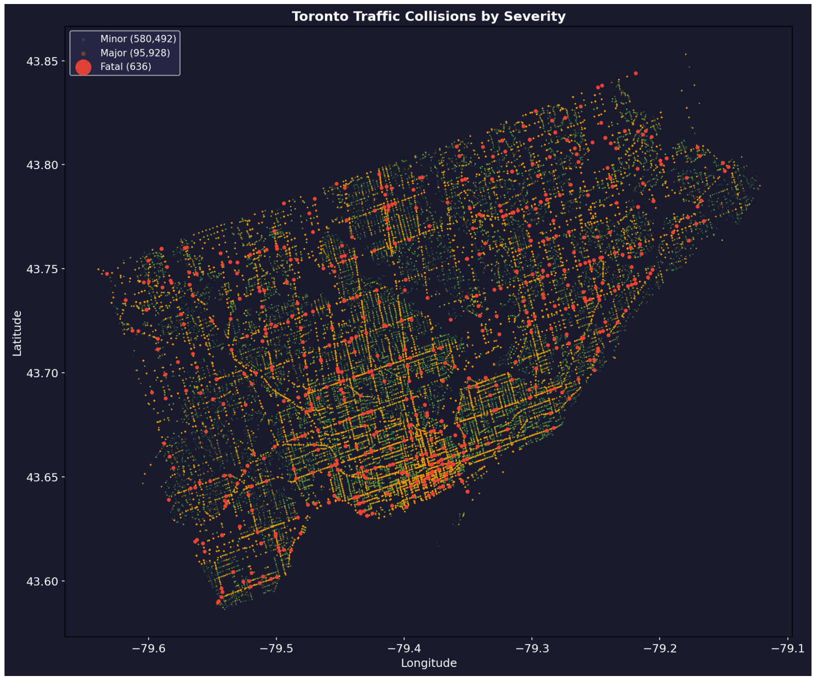

Figure 11: Toronto Traffic Collisions by Severity (GPS Plot).

DBSCAN clustering with `eps` = 300 m and `min_samples` = 3 identified 36 distinct fatal collision hotspot zones across Toronto, leaving 499 fatal collisions as noise points (i.e. spatially isolated, not part of any density-connected cluster of three or more fatal collisions within 300 m). Table 8 reports the ten largest clusters by point count.

Table 8: Ten largest DBSCAN fatal-collision clusters by point count.

Cluster ID	13	3	24	1	17	9	15	7	18	2
Points	7	6	6	5	5	5	5	5	4	4

This cluster-level view is a distinct, complementary metric to the neighbourhood-level count reported in Section 3.6 (Figure 4): the neighbourhood count sums all fatal collisions within a named Toronto neighbourhood boundary regardless of spatial proximity within that boundary, whereas the DBSCAN clusters are density-based groupings of fatal collisions within 300 m of one another, independent of neighbourhood boundaries. No neighbourhood-to-cluster mapping was performed, so the two findings are reported separately rather than merged into a single ranked list. Key geospatial findings:

- By named neighbourhood, Wexford/Maryvale leads with 18 fatal collisions, followed by West Humber-Clairville (17), South Parkdale (14), and York University Heights (13).
- By density-based cluster, the largest single cluster (ID 13) contains 7 fatal collisions in close spatial proximity; the ten largest clusters range from 4 to 7 points each, indicating that even the densest fatal-collision hotspots in Toronto are small in absolute terms.
- The 499 noise points (78.5% of all 636 fatal collisions) substantially outnumber the 137 clustered fatal collisions, indicating that most fatal collisions in this dataset are spatially isolated events rather than concentrated at a small number of recurring hotspot locations. This should temper the practical framing of hotspot-based intervention: while the 36 identified clusters are legitimate priority locations, they do not capture the majority of Toronto's fatal collisions.
- Hotspot zones concentrate along major arterial corridors in the inner suburbs. Arterial roads of this type are commonly associated with higher posted speed limits and less developed pedestrian crossing infrastructure, but speed and crossing infrastructure were not directly measured in this dataset, so this is offered as a plausible contributing factor rather than an established cause.
- The downtown core, despite having the highest total collision density, does not feature among the top fatal hotspot zones. Lower typical vehicle speeds and more developed pedestrian infrastructure downtown are plausible contributing explanations, but neither was directly measured here, so this pattern is reported descriptively rather than as a tested causal claim.
- **GPS-missingness limitation for this analysis:** the invalid-GPS rate checked in Section 3.4 differs meaningfully by severity (16.91% Minor, 12.56% Major, 4.50% Fatal), and since DBSCAN clusters only records with valid coordinates, this differential missingness directly affects which fatal collisions are eligible to appear in this section's clustering. The direction is favourable Fatal collisions are the class *least* likely to be dropped, so only 30 of 666 real Fatal records (4.50%) are excluded here, a smaller proportional loss than for Minor or Major but this check only confirms the missingness rate differs by class overall, not that it is spatially random *within* the Fatal class; if certain areas or investigation types systematically produce invalid GPS more often, the 36 identified hotspot zones could under-represent those areas specifically. This was not tested directly and is noted as an open limitation of the geospatial component.

6 Discussion

This study set out to answer two questions: whether Toronto collision severity can be predicted from time, location, weather, and roadway-context features alone, and whether any resulting predictions can be explained in a policy-relevant way. This section interprets the results reported in Section 5 against those two questions; it does not restate them. Table 9 consolidates these findings across all five experiments and the explainability and geospatial analyses, as a reference point for the interpretation that follows.

Three results dominate the overall picture. First, the outcome-information audit removed four involvement variables that were strongly associated with the recorded severity outcome, and the corrected models subsequently produced much weaker Fatal-class performance than the earlier pipeline. This supports the interpretation that the original results were inflated by methodological problems, while not implying that every remaining modelling limitation has been eliminated. Second, correcting the SMOTE-NC treatment of one-hot temporal groups removed invalid categorical combinations and materially changed the model's SHAP ranking. Third, after these corrections, all five model families still show practically weak Fatal-class discrimination in absolute terms. XGBoost leads Macro-F1 and Accuracy, whereas Random Forest leads Fatal-class PR-AUC. The consistency of weak Fatal-specific performance across different model families supports the interpretation of limited Fatal-class separability under the current predictors and modelling setup, but it does not completely rule out improvement from richer predictors, alternative model classes, or different evaluation procedures. The geospatial analysis is complementary rather than predictive: DBSCAN identifies descriptive concentrations of confirmed Fatal collisions, while most Fatal observations remain outside the identified clusters. The methodological contribution of the project is therefore the transparent auditing and correction of the modelling pipeline, together with a cautious evaluation of what the available data can and cannot support.

Table 9: Consolidated summary of experimental and analytical findings.

Section	Key Finding
EDA Severity	85.7% Minor, 14.2% Major, 0.1% Fatal. Extreme imbalance motivates the balancing strategy (Section 4.2).
EDA Spatial	Wexford/Maryvale leads with 18 Fatal collisions; descriptive Fatal-collision concentrations are observed along several inner-suburban arterial corridors. *(Stable not leakage- or encoding-affected.)*
Leakage Audit	`PEDESTRIAN`, `MOTORCYCLE`, `PASSENGER`, `BICYCLE` found to carry Fatal-rate ratios of 3–21$\times$ vs. overall; removed along with two derived features prior to modelling.
Encoding Fix	Day-of-week, season, and hour-category one-hot groups were being resampled as independent binary columns rather than single categorical variables, corrupting 5.8–34.8% of resampled rows; fixed by collapsing to a single categorical column before SMOTE-NC and re-expanding after.
Exp 1 Models (CV)	XGBoost and Gradient Boosting lead F1 Macro (0.338, 0.334), meaningfully ahead of Random Forest, Neural Network, and Logistic Regression (0.306, 0.287, 0.267). PR-AUC (Fatal) 0.0010–0.0017 for all five (close in absolute terms to the prevalence-based reference level of approximately 0.0009). XGBoost leads F1 Macro and Accuracy; Random Forest leads Fatal-class PR-AUC (0.0017) no single model is best on every metric.

Table 9 (continued)

Section	Key Finding
Exp 2 Balancing	Four of five strategies (SMOTE-NC, ADASYN, SMOTETomek, No Resampling) achieve exactly 0% Fatal recall on the validation split. Only Random Undersampling recovers Fatal recall (41.1%), at the cost of overall accuracy collapsing to 37.3%. SMOTE-NC remains best by F1 Macro among majority-preserving strategies.
Exp 3 Threshold	Lowering the threshold from 0.50 to 0.05 raises Fatal recall from 0% to 59.0%, at a 38.5-point accuracy cost and 0.11% Fatal precision ($\sim$900 false flags per true catch). Does not support an operational deployment recommendation.
Exp 4 Tuning	Internal search CV score (0.93 XGBoost) is not reproduced on real validation data: tuning decreases XGBoost F1 Macro (0.337 $\rightarrow$ 0.318) with no recall change, and increases Random Forest's F1 Macro (0.308 $\rightarrow$ 0.339) while its Fatal recall collapses (17.9% $\rightarrow$ 1.1%).
Exp 5 CV	Fold-to-fold standard deviation is small (e.g. $\pm$0.001 F1 Macro for XGBoost) this demonstrates fold-to-fold stability, not proof of real-world generalization, and does not change the practically weak Fatal-class finding.
SHAP Fatal	With the one-hot fix applied, individual day-of-week indicators are no longer dominant; the ranking is now led by `is_weekend`, `OCC_YEAR`, `OCC_MONTH`, and location features (`HOOD_158`, `neighbourhood_freq`) a more conventional, defensible structure than the pre-fix, day-of-week-heavy ranking, though the practically weak PR-AUC still limits how much causal weight this ranking should be given.
Geospatial	36 DBSCAN hotspot zones; top 3: Wexford/Maryvale (18), West Humber-Clairville (17), South Parkdale (14). *(Stable.)*

6.1 Why Fatal-Class Prediction Remains Difficult

Several results reported in Section 5, read together, point to a consistent explanation for why Fatal-class prediction remains difficult even after both corrections. First, XGBoost (Section 5.1) leads all five models on both accuracy (0.814) and F1 Macro (0.338), yet its confusion matrix (Appendix A.8) shows it correctly identifies **zero** of 95 real Fatal collisions on the validation split, calling 86 of them Minor and 9 Major. This is the clearest illustration in the study of why accuracy and F1 Macro, although both improved for XGBoost relative to the other four models, are still not sufficient on their own to establish genuine Fatal-class ability: a model can lead the field on both metrics while still detecting no Fatal cases at all, because Fatal collisions are too rare (0.1% of records) to move either metric meaningfully. This is precisely why PR-AUC (Fatal), not accuracy or F1 Macro, is the metric that answers this study's central research question (Section 4.4), and on that metric XGBoost's 0.0012 sits within 0.0003 of the majority-class baseline's 0.0009 (Section 5.1a).

Second, the balancing-strategy comparison (Section 5.2) shows that only Random Undersampling recovers non-trivial Fatal recall (41.1%), while SMOTE-NC, ADASYN, SMOTE-Tomek, and class-weighting all achieve exactly 0.0% Fatal recall on the validation split. The mechanism is straightforward: Random Undersampling is the only strategy that removes the overwhelming majority of Minor examples from the training set (from 473,939 down to 1,335 records), which forces the model to weight the comparatively few real Fatal examples much more heavily during training, at the direct cost of no longer being able to distinguish Minor from Major reliably. The SMOTE-family strategies, by contrast, produce exactly equal class sizes in the training set (406,344 records per class for SMOTE-NC) yet still achieve 0.0% Fatal recall on the validation split. This pattern is consistent with limited Fatal-class separability under the available predictors and current modelling setup. The synthetic Fatal observations generated from existing minority examples do not necessarily form a region that is cleanly separable from real Minor and Major observations. Balancing the training-set class *proportions* does not manufacture separating information that is not present in the underlying features, so at prediction time, applied to the real, highly imbalanced validation distribution, the resulting decision boundary still defaults to the majority classes. Random Undersampling instead achieves non-trivial Fatal recall by discarding the great majority of Minor training examples outright, trading away Minor/Major discrimination for Fatal sensitivity, rather than by adding usable Fatal-class signal. This is a direct illustration of the precision-recall trade-off inherent to extreme class imbalance: no balancing strategy tested here achieves both preserved majority-class performance and meaningful Fatal recall simultaneously, because balancing the training distribution alone does not guarantee that Fatal and non-Fatal observations become well separated in the available feature space.

Third, the hyperparameter-tuning results (Section 5.4) show the same tension from a different angle: tuning Random Forest for F1 Macro improved that metric ($0.308 \rightarrow 0.339$) precisely by *sacrificing* Fatal recall ($17.9\% \rightarrow 1.1\%$), and the search process's own internal score (0.79 for Random Forest, 0.93 for XGBoost) bore no relationship to either model's real validation performance. A search optimizing for an aggregate metric on resampled internal folds has no way to know that the practically important outcome is Fatal-class recall specifically, and will happily trade it away if doing so improves the metric it was told to optimize.

Fourth, the SHAP results (Section 5.6), now recomputed against the corrected model, are more internally consistent than before: the pre-fix ranking's domination by six individual day-of-week indicators is gone, replaced by a more conventional ranking led by `is_weekend`, `OCC_YEAR`, `OCC_MONTH`, and location features. This is genuine evidence that the one-hot/SMOTE-NC fix mattered substantively, not just numerically the previous ranking was very likely explaining the encoding artifact itself rather than the collisions. This improvement should not, however, be read as SHAP now identifying reliable Fatal-class risk factors: given the underlying practically weak Fatal-class PR-AUC (Section 5.1), the model being explained has very little genuine ability to distinguish Fatal collisions in the first place, and explaining a weak model's internal reasoning is itself of limited substantive value, independent of whether the explanation is computed correctly. The corrected ranking is a more defensible description of what the model is doing with a real Fatal sample of only 636 collisions than the pre-fix ranking was, but it is a description of model behaviour, not a validated or causally-interpretable set of risk factors, and this study avoids drawing policy or enforcement conclusions from it on that basis (Section 6.2).

Fifth, the geospatial findings (Section 5.7) sit apart from this picture precisely because they do not depend on predicting Fatal outcomes from pre-collision features at all; DBSCAN clusters collisions that have already occurred and were already labelled Fatal, using only their spatial coordinates. This is why the geospatial results remain stable and interpretable while the classification results do not: spatial clustering of confirmed outcomes is a fundamentally easier task than predicting an outcome that has not yet occurred from a limited feature set.

6.2 Practical Implications for Road Safety

Given the corrected results in Section 5, the practical implications of this study are narrower, but more honestly grounded, than the original draft claimed:

- **The geospatial component provides a descriptive spatial prioritization layer.** The DBSCAN clusters (Section 5.7) identify locations where multiple confirmed Fatal collisions occurred within short distances. Because roadway speed, crossing infrastructure, traffic exposure, and related characteristics were not directly measured, the clusters are not interpreted as evidence for any specific intervention on their own.
- **The predictive model, as currently specified, is not ready for operational deployment for Fatal-collision flagging.** With PR-AUC (Fatal) of 0.0010–0.0017 across all five tested model families including both XGBoost, the leading model by F1 Macro and Accuracy, and Random Forest, the leading model by Fatal-class PR-AUC Fatal-class discrimination remains practically weak in absolute terms across the evaluated models, with values close to the prevalence-based reference level of approximately 0.0009. Recommending threshold-based deployment is not supported by the corrected results (Section 5.3, where a 58.95% recall gain costs 38.5 accuracy points) and this study makes none.
- **The SHAP risk-factor ranking is now more defensible, but should still not be used on its own to guide targeted enforcement campaigns.** The corrected ranking (`is_weekend`, `OCC_YEAR`, `OCC_MONTH`, location features) is a substantial improvement over the pre-fix, day-of-week-fragmented ranking, and `is_weekend`'s directional effect (weekday collisions carrying more Fatal-class signal than weekend ones) is a specific, testable pattern worth further investigation. Given the practically weak Fatal-class PR-AUC, however, this study still stops short of recommending it as a basis for operational decisions on its own.
- **Hyperparameter tuning should not be adopted by default.** Section 5.4 shows a concrete case (Random Forest) where tuning for F1 Macro improved that metric while cutting Fatal recall by more than 16 percentage points, and a second case (XGBoost) where tuning simply made both metrics worse. Any future use of this pipeline should evaluate a tuned model's Fatal-class recall specifically, not just its aggregate score, before adopting it.

- **The main policy-relevant contribution of this study is methodological**: it demonstrates, with a like-for-like comparison across five model families, that a widely-used approach to this class of problem (involvement-flag features, uncorrected SMOTE, and unseparated validation/test data) can produce misleadingly strong-looking results, and that removing the leakage, fixing the categorical encoding, and separating validation from test data reveals substantially less real predictive signal in the remaining features than an uncorrected approach would suggest. This is directly relevant to any group evaluating similar collision-severity models before operational use.

6.3 Comparison with Literature

Table 10 situates this study's corrected results against the five most comparable studies from Section 2. Unlike the original draft, this comparison leads with F1 Macro and Fatal-class PR-AUC the metrics this study's own methodology (Section 4.4) argues are appropriate for a severely imbalanced, safety-critical target rather than accuracy, and does not claim this study achieves the best performance of the group. It does not: on any metric reported by the comparison studies, this study's corrected results are weaker, not stronger.

Table 10: Comparison with related literature. Reported metrics differ across studies (accuracy, ROC-AUC) since few report PR-AUC or F1 Macro directly; this study's corrected F1 Macro and PR-AUC (Fatal) are given for reference rather than as a directly comparable like-for-like figure. As noted below the table, two entries required correction during final review, and all figures in this table should be treated as this study's reading of each source and verified independently before further reuse.

Study	Dataset	Best Model	Reported Performance
Parsa et al. [6]	US highway crash	XGBoost + SHAP	Real-time crash detection task (not severity classification); performance figure not independently confirmed for this report
Sameen & Pradhan [8]	Malaysian highways	Recurrent Neural Network†	Reported performance not independently confirmed for this report
Chen & Wang [2]	US national crash	Random Forest	Accuracy = 0.84
Sheykhfard et al. [9]	Iranian rural roads	Gradient Boosting	Accuracy = 0.82
Al-Ghamdi [1]	Saudi Arabia urban	Logistic Regression	Accuracy = 0.78
This Study (corrected)	Toronto urban (677k)	XGBoost	F1 Macro = 0.338; PR-AUC (Fatal) = 0.0012

†This table originally listed the Sameen and Pradhan model as a Multi-Layer Perceptron and

the Parsa et al. result as an AUC of 0.87 for a SMOTE-oversampled severity-classification task; a final review found these did not match the cited papers' actual models/tasks as accessible during this review, so both entries were corrected here. Given this, the Chen & Wang, Sheykhfard et al., and Al-Ghamdi accuracy figures above have not been re-verified against the primary sources and are presented with the same caveat.

Direct comparison is complicated by several factors: the reviewed studies use binary (injury/no-injury or KSI/non-KSI) rather than three-class targets, several report on class imbalances of 2–5% rather than the 0.1% Fatal rate studied here, and none report PR-AUC, which this study's own methodology argues is the more appropriate metric at this degree of imbalance so the comparison in Table 10 should be read as contextual rather than a rigorous like-for-like ranking. With that caveat, this study's corrected results are weaker than every comparison study on their own reported metrics, and this study does not claim otherwise. This is consistent with this study's central finding: once involvement-flag features that are plausibly leakage-affected are removed, and class imbalance is handled with a categorical-aware method, the remaining genuinely pre-collision features (time, weather, location, year) carry substantially less Fatal-class signal than the pre-correction numbers, or the comparison literature's typically less extreme class imbalances, would suggest. None of the five reviewed studies report a comparable leakage audit of their own engineered involvement features, which this study's Section 2 gap statement identifies as a specific limitation of the existing literature that this study addresses directly; the corrected results here suggest that gap may be substantively important, not merely a methodological formality, for studies reporting strong Fatal/KSI-class performance from similar involvement-flag features.

The methodological contributions of this study the systematic leakage audit and correction, the categorical-aware SMOTE-NC balancing strategy, the full-dataset cross-validated model comparison across five model families, and the integration of SHAP explainability with DBSCAN geospatial clustering on dense urban Canadian collision data merged with real hourly Canadian weather observations are not present jointly in any of the five reviewed studies, independent of the final performance figures.

7 Conclusion

This study developed and evaluated a machine learning framework to predict Toronto traffic collision severity from time, location, weather, and roadway-context features, and to explain the resulting predictions through SHAP and complement them with DBSCAN-based geospatial clustering. In doing so, it identified and corrected two methodological issues: a data-leakage problem in four involvement-flag features, and a categorical-encoding error in which SMOTE-NC was resampling already one-hot-encoded temporal features as independent binary columns rather than as single categorical variables, corrupting 5.8–34.8% of resampled training rows across the affected groups. It also separated the validation split used for model/strategy/threshold selection from the test split reserved solely for the SHAP explainability analysis, and evaluated five classifier families under a leakage-free, correctly-encoded, full-dataset, 5-fold cross-validation design. The headline performance figures reported throughout this study (Table 3) come from this full-dataset cross-validation, not from the held-out test split, which is reserved for the SHAP analysis alone (Section 4.1).

The corrected results answer the two research questions with an important distinction between overall and Fatal-specific performance. XGBoost achieved the highest F1 Macro (0.3381) and Accuracy and is therefore the strongest overall three-class model under the chosen aggregate criterion. Random Forest achieved the highest Fatal-class PR-AUC (0.0017), making it the strongest evaluated model on the Fatal-specific discrimination metric. Nevertheless, Fatal PR-AUC remained between 0.0010 and 0.0017 across all five model families, values that are close in absolute terms to the approximate Fatal-prevalence reference level of 0.0009 and therefore indicate practically weak Fatal-class discrimination. The consistency of this pattern across several model families supports the interpretation that the current predictors provide limited Fatal-class separability under the present modelling setup, while not ruling out improvements from alternative predictors, model classes, or evaluation designs. SHAP is consequently interpreted as an explanation of model behaviour rather than as causal evidence. DBSCAN identified 36 descriptive Fatal-collision clusters, but 499 of 636 Fatal observations were classified as spatial noise, so the clusters should be treated as a supplementary spatial prioritization layer rather than as a comprehensive Fatal-risk map. Overall, the study's contribution is both analytical and methodological: it demonstrates the importance of outcome-information auditing, correct categorical resampling, explicit metric hierarchy, and cautious interpretation in a severely imbalanced transportation-safety problem.

8 Limitations and Threats to Validity

- **Fatal-class discrimination remains practically weak under the corrected feature set and modelling setup.** PR-AUC (Fatal) of 0.0010–0.0017 across all five tested model families (Section 5.1) is close in absolute terms to the approximate Fatal-prevalence reference level of 0.0009 (Section 5.1a). A formal test of statistical distinguishability between these values was not performed, so this is reported as practically weak Fatal-class performance rather than as a claim that the difference from the baseline is statistically indistinguishable from zero. This supports the interpretation that time, weather, location, and neighbourhood-frequency features provide limited Fatal-class separability under the current modelling setup; it does not rule out improvements from richer predictors or alternative modelling approaches. This is the central limitation of the current feature set, not a limitation of any one model choice, and it holds under both the leakage correction and the subsequent one-hot/SMOTE-NC encoding fix.
- The Fatal class contains only 636 genuine records; despite SMOTE-NC generating synthetic samples, the model's ability to learn robust and generalizable Fatal patterns is inherently constrained by this small real sample size, independent of feature quality or encoding correctness.

- **The headline cross-validation is not a strictly independent, nested-CV estimate relative to the validation-split-based selection decisions.** As stated explicitly in Section 4.1, the balancing strategy (SMOTE-NC), classification threshold, and tuned-vs-default hyperparameter choice were all selected using the validation split, while the headline model comparison (Table 3) re-partitions the *full* dataset including those same validation records into fresh cross-validation folds. A fully nested cross-validation was not performed given the computational cost of refitting SMOTE-NC and five classifiers per outer fold; the resulting figures should be read as a large, stable re-evaluation of an already-selected configuration rather than a fully held-out generalization estimate.

- **ADASYN and SMOTETomek, used only in the balancing-strategy comparison (Table 4), retain the categorical-interpolation limitation that motivated the SMOTE-NC fix.** Neither has a categorical-aware mode in `imbalanced-learn`, so both were applied to the fully one-hot-expanded training data (Section 4.2) and can still produce the same impossible fractional/multi-hot combinations that SMOTE-NC's collapse/expand fix specifically avoids; this is a known, disclosed limitation of those two comparison strategies, not an oversight, and is a further reason SMOTE-NC not ADASYN or SMOTETomek is this study's primary balancing method.

- **Hyperparameter tuning, as currently configured, cannot be trusted without a held-out check.** Section 5.4 shows `RandomizedSearchCV`'s internal cross-validated score (0.93 for XGBoost, 0.79 for Random Forest) bears no relationship to real validation performance, and in Random Forest's case tuning traded a 16.8-point Fatal recall loss for a modest F1 Macro gain. Any future use of this pipeline must evaluate tuned models against a genuinely held-out split before adopting tuned parameters, not the search's own reported score.

- Wind speed and visibility data were entirely unavailable from Toronto City Station ID 31688 (100% null values), limiting the weather feature set to temperature, humidity, dew point, and precipitation.

- A single central weather station cannot capture hyperlocal weather conditions across a geographically large city.

- The dataset includes only officially reported collisions; unreported, predominantly minor, incidents are absent, potentially introducing reporting bias into the Minor class.

- Four involvement-flag features were found to be leakage-affected and removed via a targeted audit driven by their unusually high SHAP rankings in the original model; it is possible that other features in the dataset carry subtler, undetected leakage that did not produce as conspicuous a signal and was therefore not identified by this audit.

- **The `neighbourhood_freq` feature was frequency-encoded on the full dataset before the train/validation/test split.** It is computed as each neighbourhood's share of all 677,056 records, which means validation- and test-set rows contribute to the frequency value seen during training. This is not target leakage, since it does not use the severity label, but it is a form of evaluation contamination: the feature's value for a given row is influenced by held-out records rather than derived from the training partition alone. A corrected implementation would compute this frequency within the training fold only. Given the very small size of this effect relative to the study's central finding (practically weak Fatal-class PR-AUC across all models), it is not expected to change the substantive conclusions, but it was not re-run and quantified before this submission and is disclosed here as an open item for future work.

- **The Random Forest hyperparameter-tuning subsample is seeded and reproducible; the SHAP explainability sample's seeding status has not been verified.** The Random Forest subsample (Table 6) is drawn via `np.random.RandomState(RANDOM_STATE)`, so results from this draw are now exactly reproducible. The SHAP explainability sample draw has not been confirmed to use the same seeded pattern; this should be checked before this pipeline's SHAP results are used for any further reporting.

- 2026 is a partial year in this dataset (4 of 12 months present, 15,467 records, 19.3% below the same four months in 2025). Records were flagged rather than excluded, since they remain valid individual collision observations. Unlike the pre-fix version of this study, `OCC_YEAR` now ranks first in the corrected global SHAP importance (Section 5.6), so this caveat is more consequential than previously stated: any interpretation of `OCC_YEAR`'s SHAP contribution should explicitly account for 2026 being incomplete, and any future year-over-year trend analysis using this dataset should adjust for the incomplete 2026 window rather than comparing raw annual totals.

- Hyperparameter tuning for Random Forest was conducted on a 200,000-record subsample due to memory constraints; while this subsample draw is now seeded (see above), the subsample itself remains a smaller, non-representative slice of the full training set.

A Supplementary Exploratory and Model Plots

The main report focuses on figures that directly support the research questions and principal findings. The following supplementary figures preserve additional exploratory and modelling outputs without overcrowding the main narrative.

A.1 GitHub Repository

All source code and additional data required to reproduce these results are available at:
https://github.com/Nishi-013/toronto-collision-severity-prediction

A.2 Monthly Collision Distribution

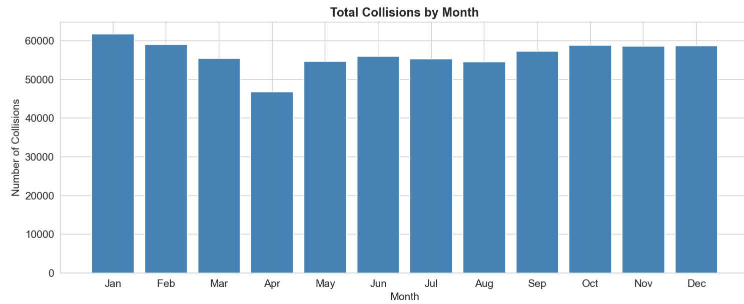

Figure 12: Total Collisions by Month.

A.3 Day-of-Week Collision Distribution

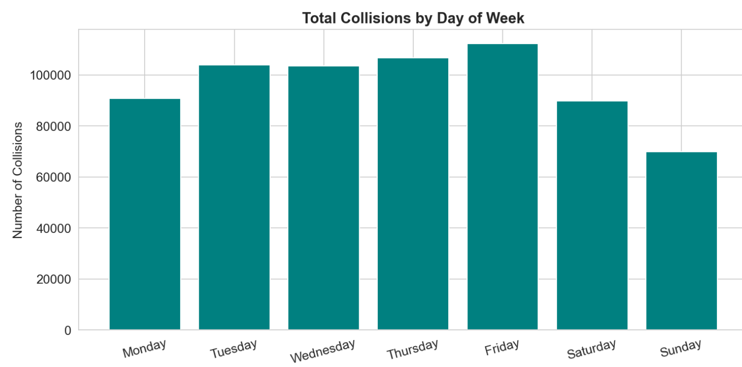

Figure 13: Total Collisions by Day of Week.

A.4 Correlation Heatmap

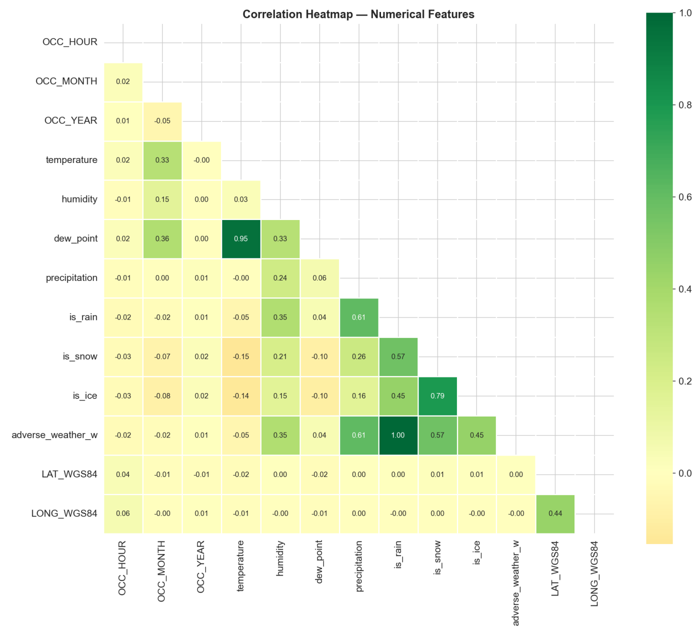

Figure 14: Pearson Correlation Matrix of Numerical Features.

A.5 Temperature Distribution by Severity

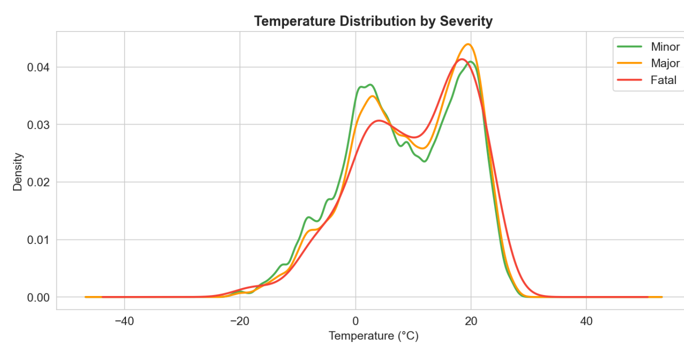

Figure 15: Temperature Distribution by Severity Class (KDE).

A.6 Single-Split Diagnostic Comparison

A separate single-split diagnostic table is not reported as its own figure in this version of the study: the primary model comparison uses the full-dataset, all-model cross-validation reported in Table 3 (Section 5.1), per the consolidation described in Section 4.6. The corresponding bar-chart and single-model confusion-matrix figures are therefore omitted here; the underlying per-model numbers are already reflected in the Table 3-adjacent text and in Table 11 below. The all-model confusion matrix grid (Figure 17, Appendix Section A.8, “All Model Confusion Matrices”) already includes XGBoost’s panel with the correct, current numbers, so the substantive result is not missing only a supplementary single-model view.

A.7 Balancing Strategy Comparison by Class

Figure 16 decomposes the balancing-strategy comparison of Section 5.2 by class-specific F1 score, supplementing the Fatal-recall view already shown in Figure 6.

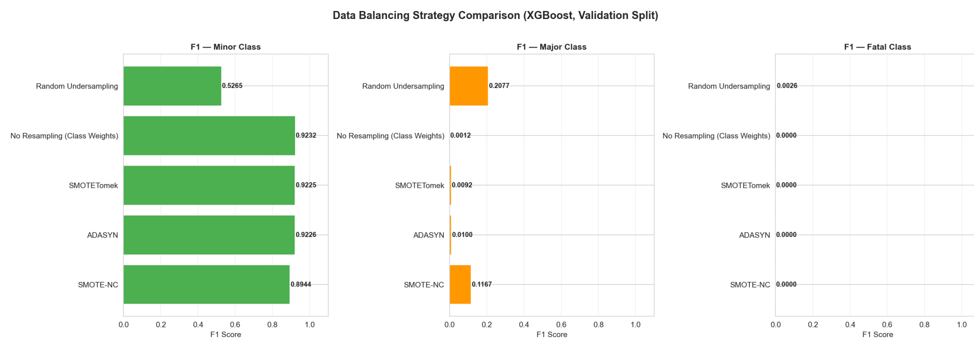

Figure 16: Per-Class F1 Score by Balancing Strategy.

A.8 All Model Confusion Matrices

Table 11 isolates the Fatal row (true label = Fatal, $n = 95$) across all five models, since this is the row of primary interest and is difficult to read precisely off the figure at this scale.

Read directly, this table complicates the simple "no model detects Fatal collisions" picture in an instructive way: of 95 real Fatal collisions in the validation split, XGBoost the model with the highest F1 Macro and Accuracy (Table 3) correctly identifies **zero**, while the other four models each catch at least a handful (Random Forest 17, Neural Network 8, Gradient Boosting 6, and, notably, Logistic Regression 23, the highest raw count of any model despite Logistic Regression trailing every other model on F1 Macro). This is a direct illustration of the precision/recall trade-off underlying this study’s metric choice (Section 4.4): Logistic Regression’s class-weighted linear decision boundary is evidently more willing to predict a minority class, catching more true Fatal cases at the cost of far more false Fatal predictions elsewhere, which is exactly why its overall F1 Macro (0.267, Table 3) is the lowest of the five models even as its raw Fatal recall here is the highest. XGBoost’s zero Fatal recall on this split, despite leading F1 Macro and Accuracy in Table 3 (though not Fatal-class PR-AUC, where Random Forest is highest), is the starkest concrete demonstration in this report of why accuracy and F1 Macro are not sufficient on their own for a problem this imbalanced, and why PR-AUC (Fatal) is reported as the primary lens on Fatal-class ability specifically (Section 4.4).

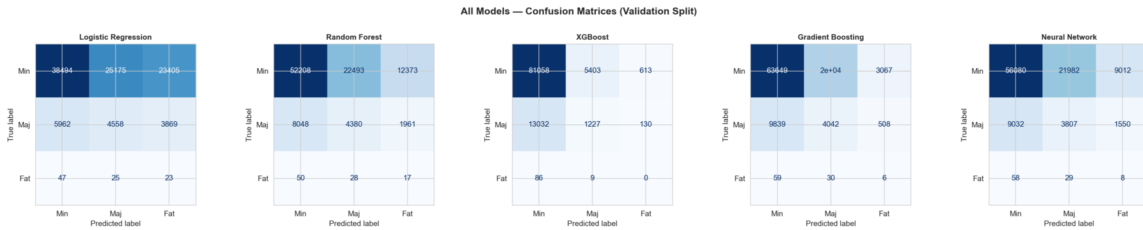

Figure 17: Confusion Matrices, All Five Models.

Table 11: Fatal-class row from each model's confusion matrix (95 real Fatal collisions in the validation split). "Correct" = predicted Fatal when true label is Fatal.

Model	Pred. Minor	Pred. Major	Pred. Fatal (correct)	Recall
Logistic Regression	47	25	**23**	**24.2%**
Random Forest	50	28	17	17.9%
Neural Network	58	29	8	8.4%
Gradient Boosting	59	30	6	6.3%
XGBoost	86	9	0	0.0%

For completeness, the full Minor and Major rows (validation split, $n = 101{,}558$) are: Logistic Regression 38,494/25,175/23,405 (Minor row) and 5,962/4,558/3,869 (Major row); Random Forest 52,208/22,493/12,373 and 8,048/4,380/1,961; XGBoost 81,058/5,403/613 and 13,032/1,227/130; Gradient Boosting 63,649/20,358/3,067 and 9,839/4,042/508; Neural Network 56,080/21,982/9,012 and 9,032/3,807/1,550 (each triple giving predicted Minor/Major/Fatal counts).

A.9 Major-Class SHAP Beeswarm

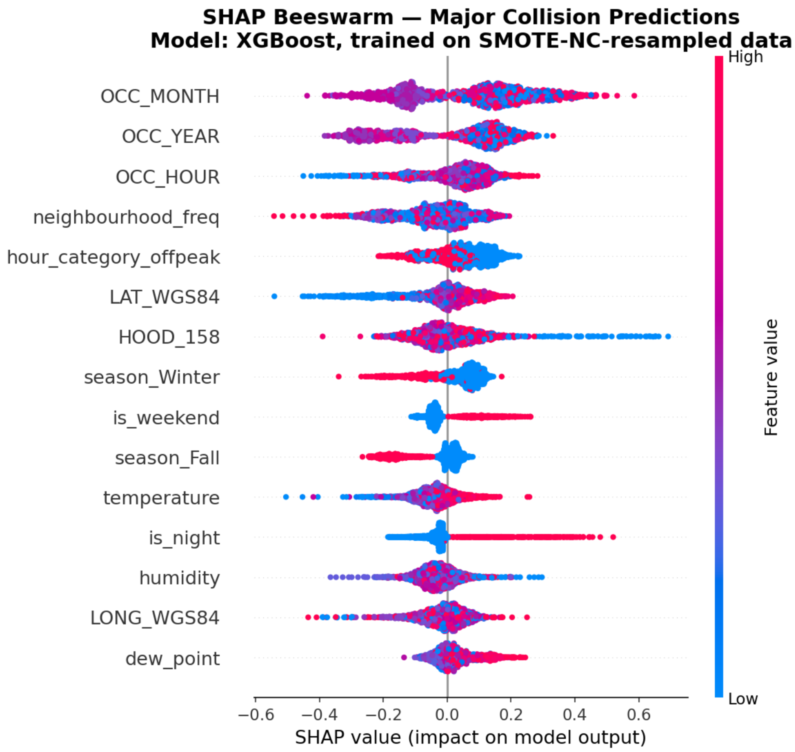

Figure 18: SHAP Beeswarm, Major Class.

A.10 SHAP Waterfall: A Single Fatal Case Explained

Figure 19 shows a SHAP waterfall decomposition for one individual, genuinely Fatal-labelled collision, illustrating at the case level how the corrected model reasons about a specific record rather than only in aggregate.

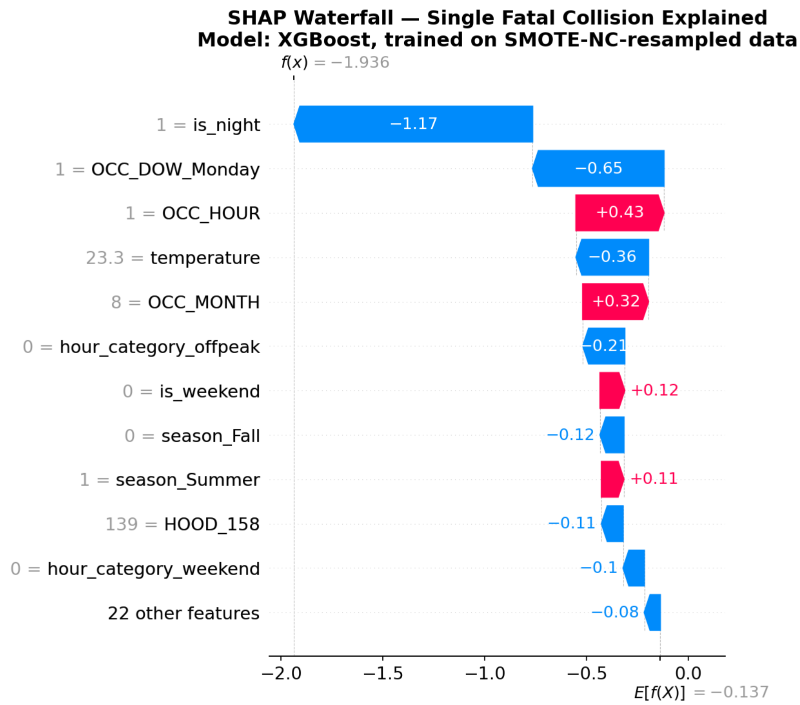

Figure 19: SHAP Waterfall: Single Fatal Case.

For this specific record, the model's output moves substantially further from Fatal than its own average prediction, driven primarily by `is_night` (-1.17) and `OCC_DOW_Monday` (-0.65), and only partially offset by positive contributions from `OCC_HOUR` ($+0.43$) and `OCC_MONTH` ($+0.32$); the remaining listed features contribute smaller amounts (-0.36 to -0.10 each), and twenty-two further features contribute a combined -0.08, individually negligible. This is a different specific case than would have been selected under the pre-fix pipeline, since the underlying model was retrained, but the qualitative pattern is the same: this case-level view is consistent with, and helps explain mechanically, the near-zero aggregate Fatal recall reported in Section 5.1 for XGBoost specifically (Table 11) for at least this real Fatal collision, the features that otherwise carry the model's Fatal-class reasoning (Section 5.6) actively push the prediction away from Fatal rather than toward it, illustrating concretely why the model fails to flag genuine Fatal cases rather than merely reporting that it does. This case-level explanation illustrates the fitted model's decision process and should not be interpreted as evidence that these factors caused the observed outcome.

A.11 Hyperparameter Tuning

Best parameters found for XGBoost and Random Forest are reported in Section 5.4 (Table 6 and accompanying discussion); no additional supplementary figure is required here.

References

[1] Al-Ghamdi, A. S. (2002). Using logistic regression to estimate the influence of accident factors on accident severity. *Accident Analysis & Prevention*, 34(6), 729–741. https://doi.org/10.1016/S0001-4575(01)00073-2

[2] Chen, J., & Wang, L. (2020). Predicting traffic accident severity using machine learning: A review of classification approaches. *Transportation Research Record*, 2674(9), 1–12. https://doi.org/10.1177/0361198120927169

[3] Toronto Police Service. (2024). *Traffic Collisions Open Data (ASR-T-TBL-001)*. Toronto Police Service Public Safety Data Portal. https://data.tps.ca/

[4] Environment and Climate Change Canada. (2024). *Historical climate data*. Government of Canada. https://climate.weather.gc.ca/

[5] Lundberg, S. M., Erion, G., Chen, H., DeGrave, A., Prutkin, J. M., Nair, B., & Lee, S. I. (2020). From local explanations to global understanding with explainable AI for trees. *Nature Machine Intelligence*, 2(1), 56–67. https://doi.org/10.1038/s42256-019-0138-9

[6] Parsa, A. B., Movahedi, A., Taghipour, H., Derrible, S., & Mohammadian, A. K. (2020). Toward safer highways, application of XGBoost and SHAP for real-time accident detection and feature analysis. *Accident Analysis & Prevention*, 136, 105405. https://doi.org/10.1016/j.aap.2019.105405

[7] Rezapour, M., Molan, A. M., & Ksaibati, K. (2020). Analyzing injury severity of motorcycle at-fault crashes using machine learning techniques, decision tree and logistic regression models. *International Journal of Transportation Science and Technology*, 9(2), 89–99. https://doi.org/10.1016/j.ijtst.2019.10.002

[8] Sameen, M. I., & Pradhan, B. (2017). Severity prediction of traffic accidents with recurrent neural networks. *Applied Sciences*, 7(6), 476. https://doi.org/10.3390/app7060476

[9] Sheykhfard, A., Haghighi, F., Nordfjærn, T., & Soltaninejad, M. (2021). Logistic regression models for rural and urban crashes on two-lane two-way roads: A case study from Iran. *IATSS Research*, 45(1), 107–114. https://doi.org/10.1016/j.iatssr.2020.06.003

[10] Xie, Z., & Yan, J. (2008). Kernel density estimation of traffic accidents in a network space. *Computers, Environment and Urban Systems*, 32(5), 396–406. https://doi.org/10.1016/j.compenvurbsys.2008.05.001